

Diversity of Study Models in Physiology



Use of animal cell models in pathophysiology



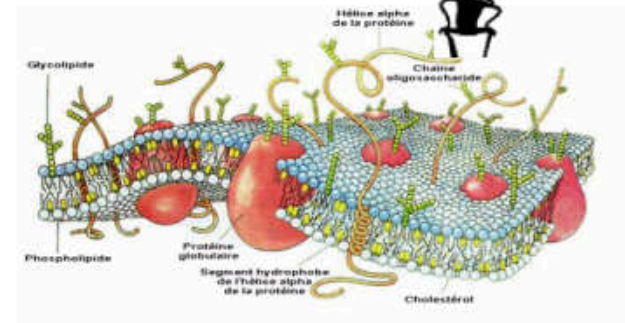
Véronique Béréziat

veronique.bereziat@sorbonne-universite.fr

Saint Antoine Research Center CRSA
Faculté de Médecine Pierre et Marie Curie Site St-Antoine

Head of the Nutrition-Quality-Health master program

Member of the teaching team for the
Human Physiology and Pathophysiology master program



Lipodystrophy, Metabolic and Hormonal Adaptations, and Aging Team

Pr Bruno Fève



Lipodystrophy, Metabolic and Hormonal Adaptations, and Aging Team

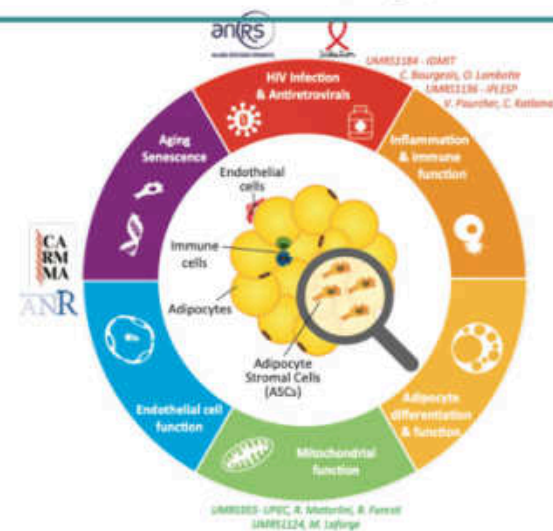
Pr Bruno Fève



HIV-associated lipodystrophies and its cardiometabolic consequences

Pr V. Béréziat
Dr C. Lagathu

veronique.bereziat@sorbonne-universite.fr
Claire.lagathu@sorbonne-universite.fr



Key figures on HIV



39 MILLIONS
People with HIV

1.3 MILLIONS
people infected with
the virus in 2023

630000
people have died
from opportunistic
infections



29.8 MILLIONS
people that had access to an antiretroviral therapy

Abnormal fat distribution in the context of HIV and antiretroviral therapy



LIPODYSTROPHIES

Heterogeneous group of genetic or acquired pathologies:

Generalized or partial lipoatrophy

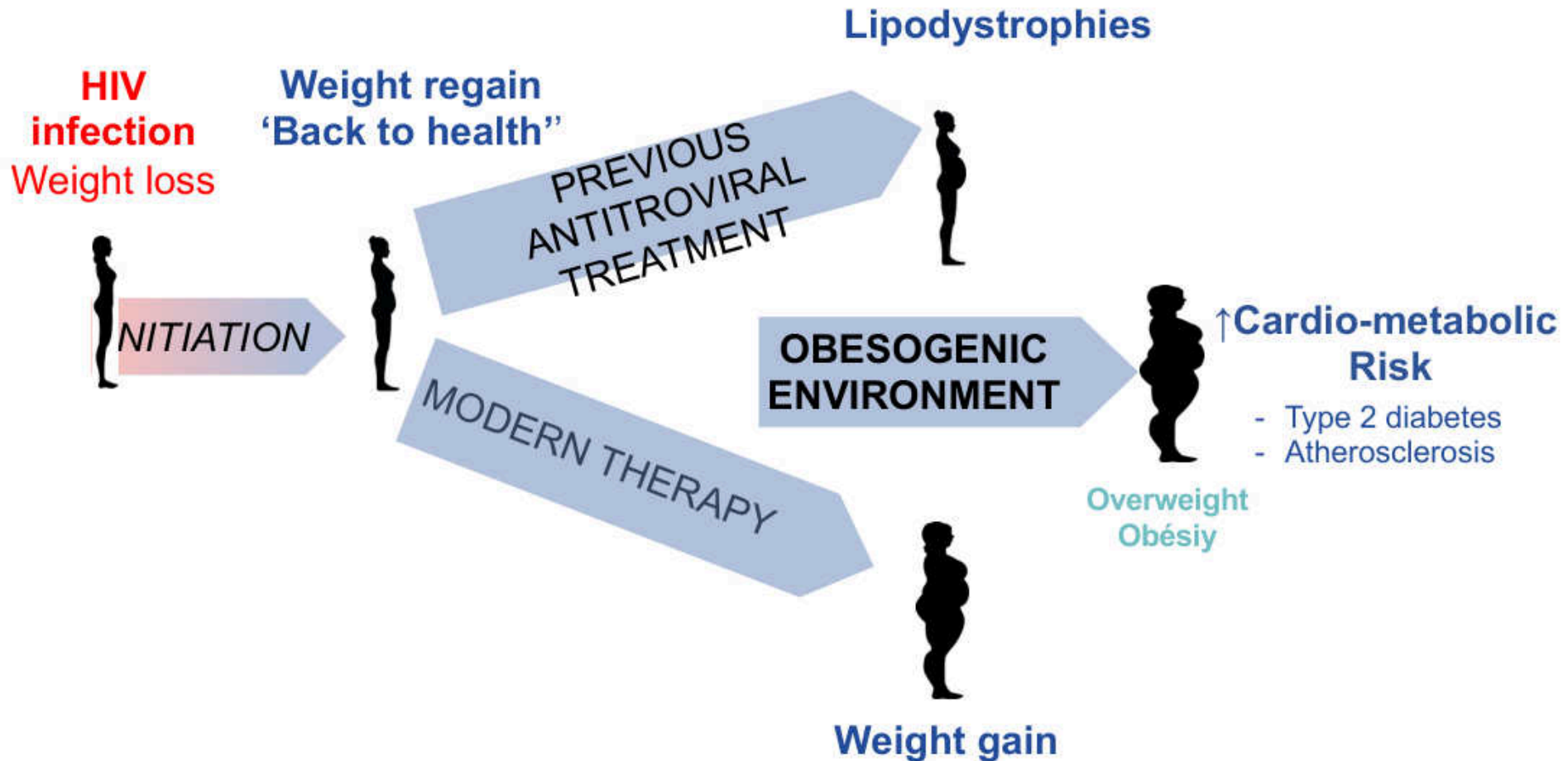
+/- Abnormal deposits of adipose tissue

- Severe insulin resistance with *acanthosis nigricans*
- Metabolic complications :
 - ✓ Diabetes
 - ✓ Dyslipidemia
 - ✓ Hepatic steatosis
- Cardiovascular diseases

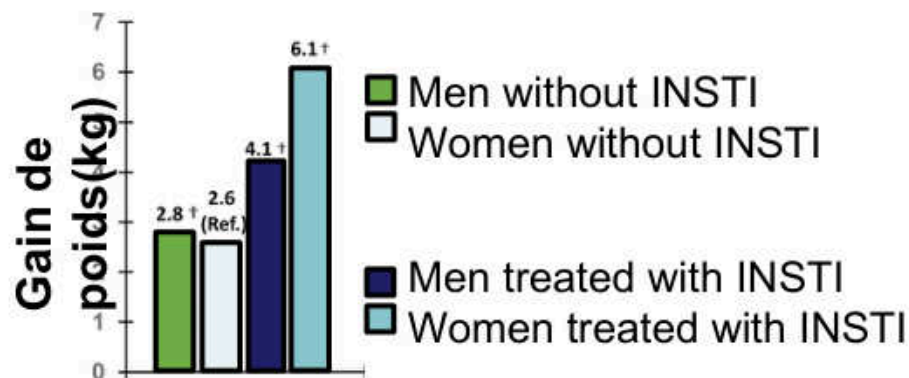


*Acanthosis
nigricans*

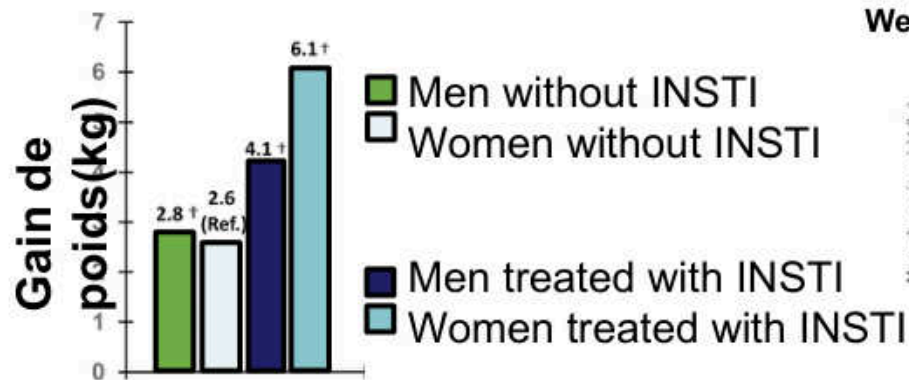
Abnormal fat distribution in the context of HIV and antiretroviral therapy



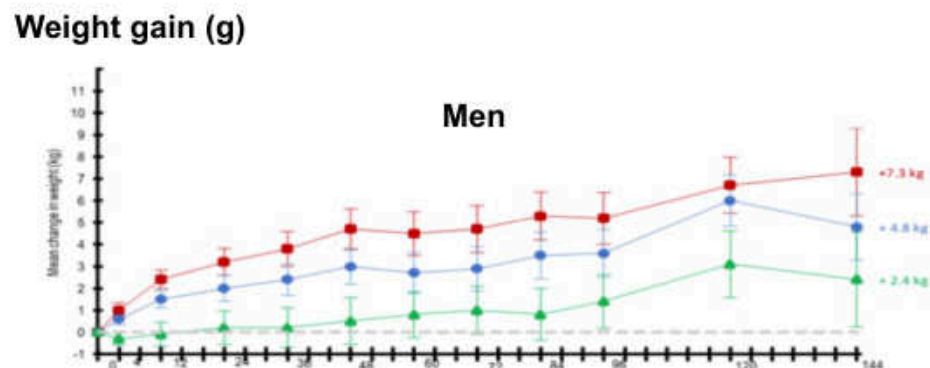
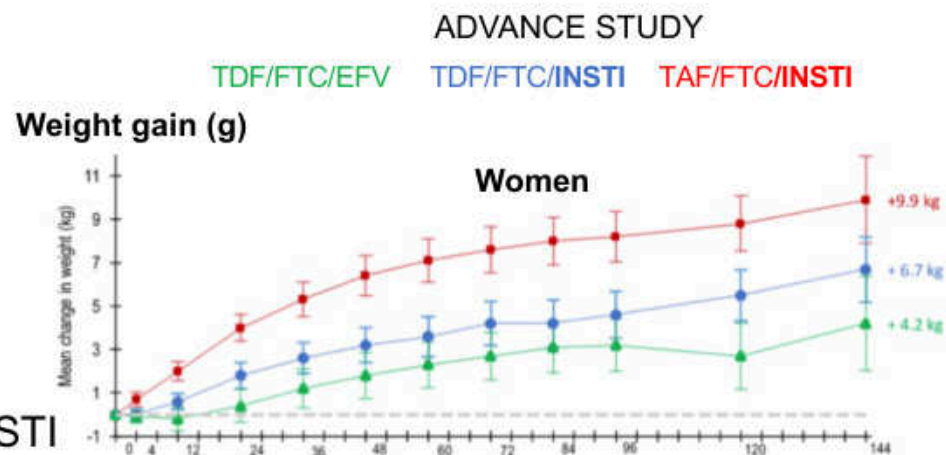
Weight gain risk factors in people with HIV treated with INSTI-containing regimen



Weight gain risk factors in people with HIV treated with INSTI-containing regimen



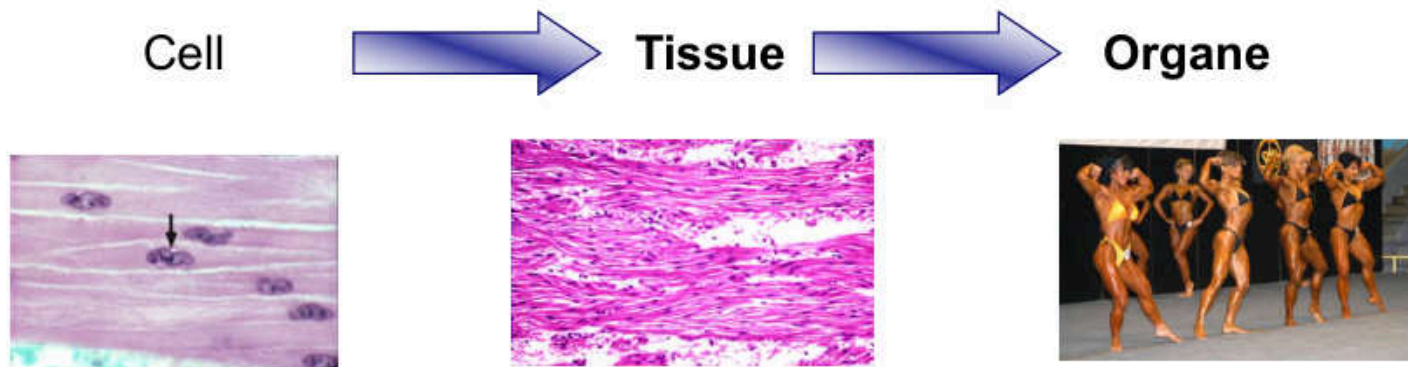
18 months after
INSTI exposure



A- Mechanisms involved in weight gain and adipose tissue damage mediated by new antiretroviral drugs in people living with HIV

I- Definition

Cellula (small room): structural, functional, and reproductive unit constituting living beings.



Cell culture: cultivate cells of various types and origins *in vitro*

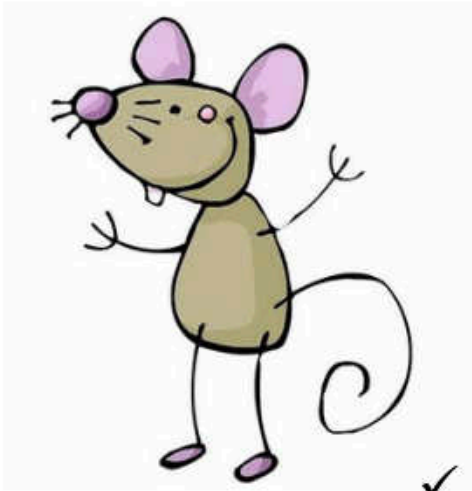
- basic/applied research
- toxicology
- karyotype analysis
- organ production
- Etc...



STUDY MODELS

Genetic or acquired Lipodystrophies

1. *IN VIVO* studies, on an entire organism



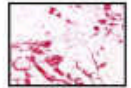
Patient Animal Model

- ✓ Study of clinical parameters
 - ✓ Functional study
- ✓ Clinical protocol: study of a new treatment

STUDY MODELS

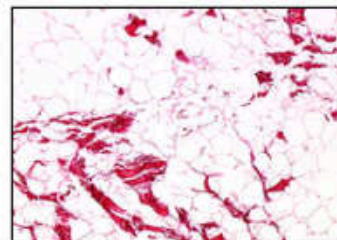
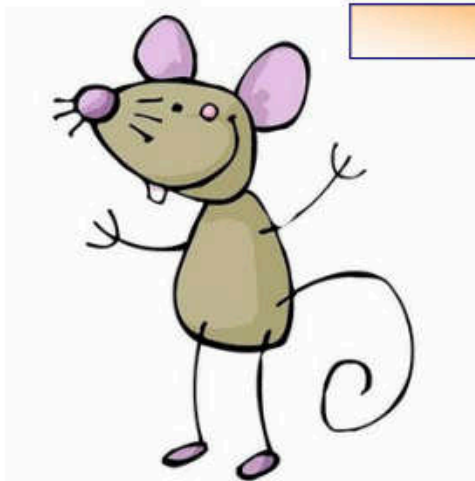
Genetic or acquired Lipodystrophies

2. *EX VIVO* studies: on an entire organ/tissue
A COMPLEX cell model



STUDY MODELS

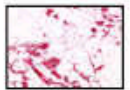
Genetic or acquired Lipodystrophies (treatment)



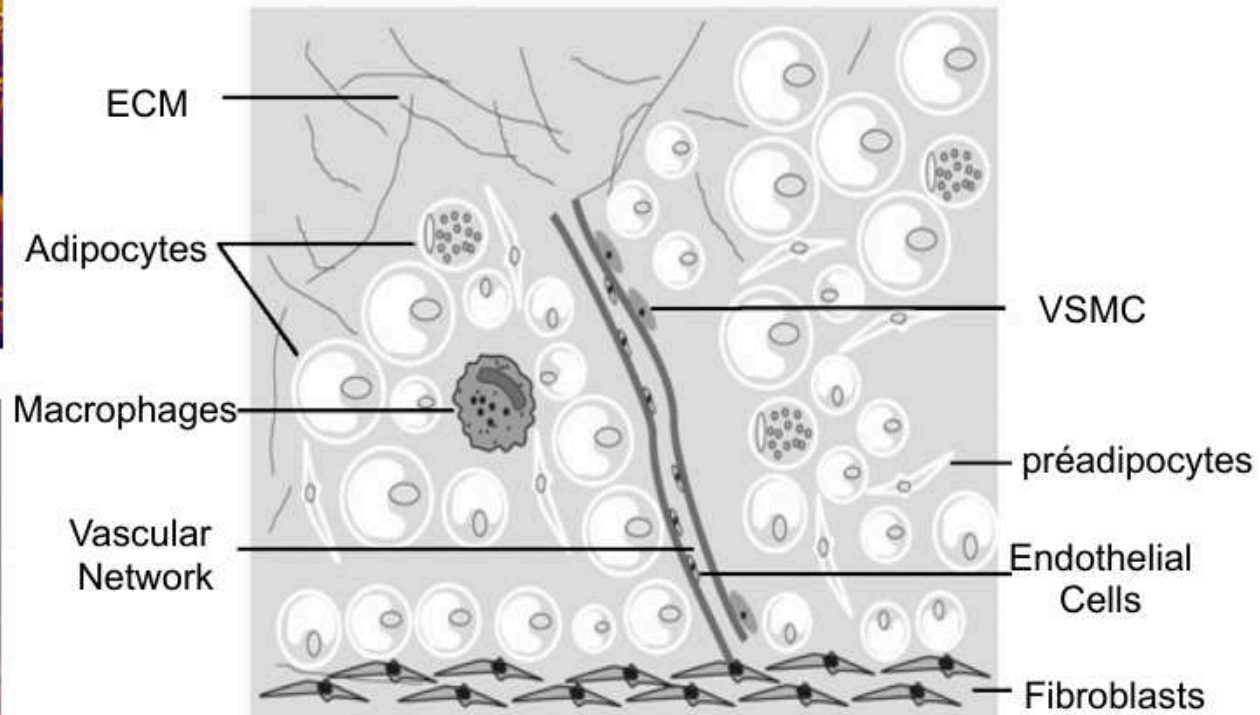
- ✓ Phenotypic exploration of biopsies:
 - Immunohistochemical analysis
 - ✓ Western Blot (protein extraction)
 - ✓ Real-time PCR (RNA extraction)

Patient
Animal model

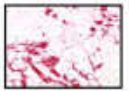
EX VIVO studies: on an entire organ/tissue



Maintain the morphological and functional differentiation of tissue fragments, organ fragments, or organ draft for at least 24 hours. These fragments are called explants.



EX VIVO studies: on an entire organ/tissue



- Keep cells alive for **at least 24 hours**
- Minimal activity**
- Relatively normal appearance**
- Maintaining their integrity**
- Reconstructing *in vitro*** the environmental and nutritional conditions provided to the cell *in vivo*

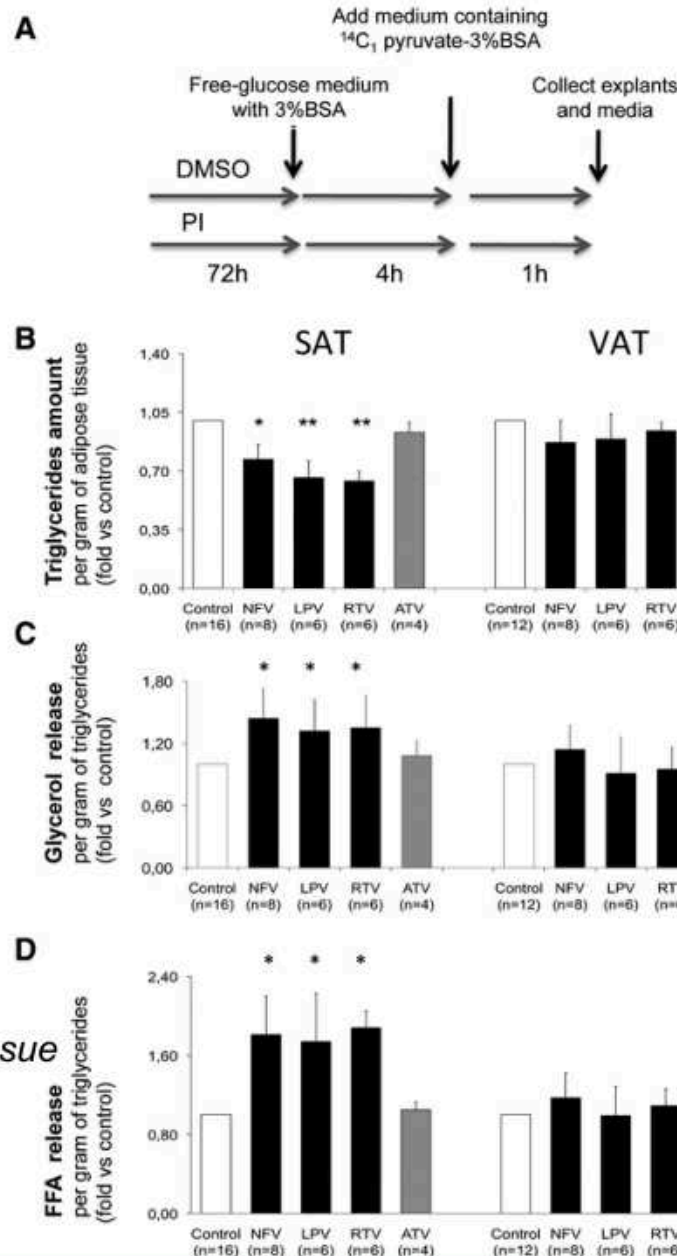
EX VIVO studies: on an entire organ/tissue



Stéphanie Leroyer
DEA UPMC, Ph.D.,
HEC Paris, Servier,
Sanofi (12 ans),
Healthcare
Professional
Engagement Manager
GSK



Leroyer S et al. *J. Lipid Res.* 2010.



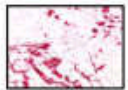
AIM: Role of HIV protease inhibitors (PIs) in the development of lipodystrophy?

RESULTS

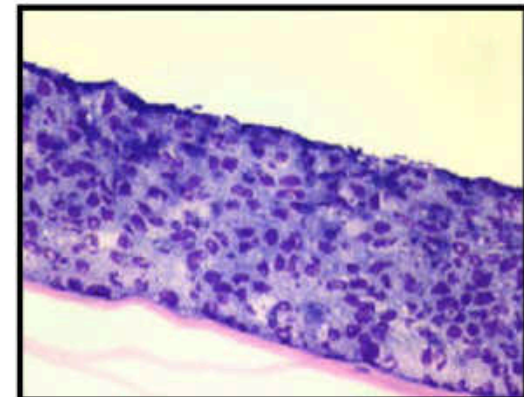
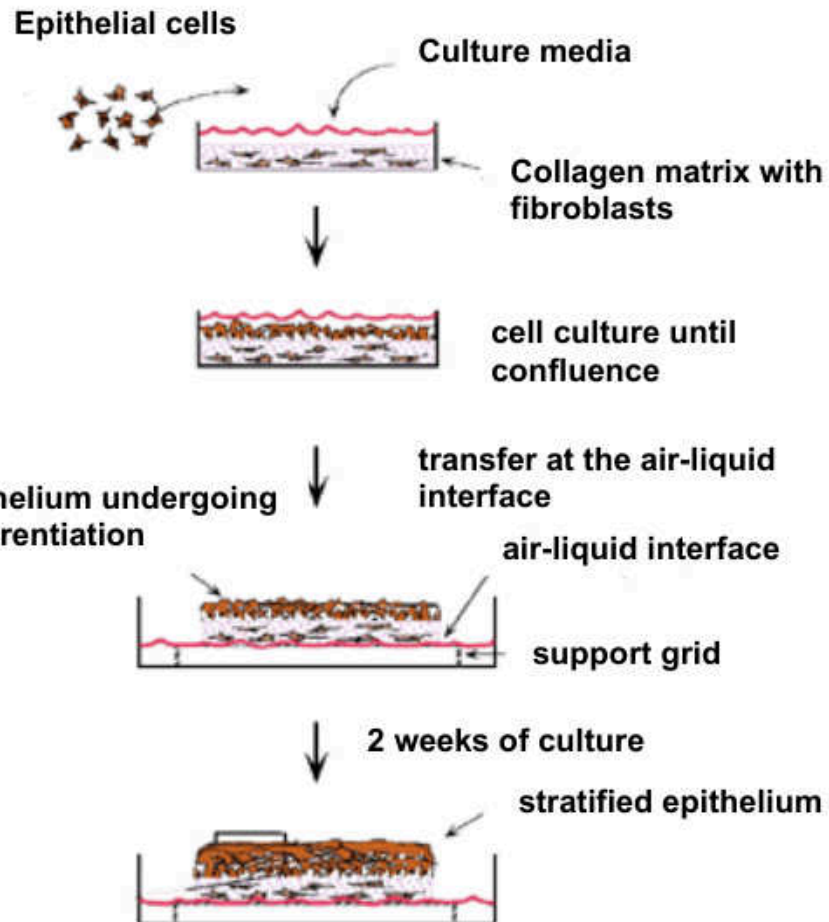
- decrease adipogenesis in SCAT (decrease triglycerides)
- increase lipolysis in SCAT (increase glycerol and FFA release)

PI: Protease Inhibitor
SCAT: Subcutaneous Adipose Tissue
VAT: Visceral Adipose Tissue

EX VIVO studies: on an entire organ/tissue

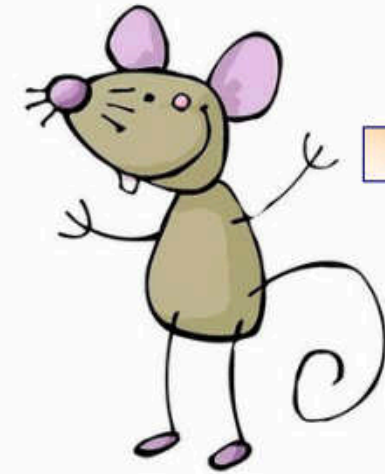


Studies on explant reconstructed *in vitro*



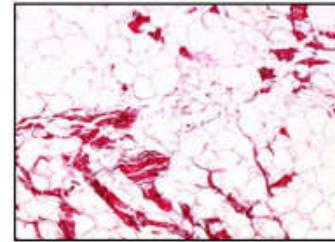
STUDY MODELS

3. *IN VITRO* studies: A SINGLE cell model



**Patient
Animal model**

« Multicellular » Tissue



**Healthy
donor**

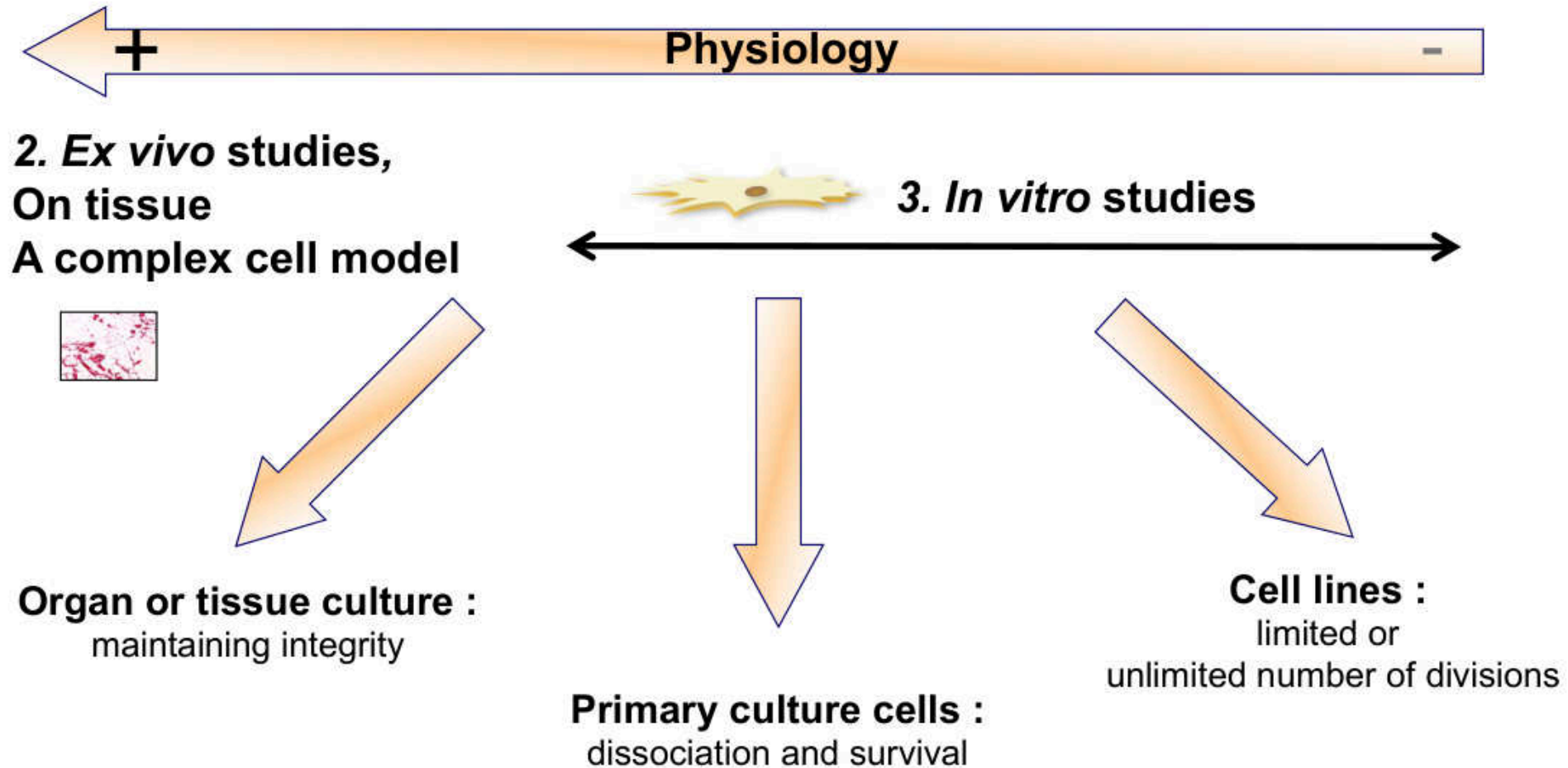
1 cell type



- ✓ Western Blot (protein extraction)
- ✓ Real-time PCR (RNA extraction)
- ✓ Functional exploration

I- Definition

STUDY MODELS



I- Scientific culture of cell culture

Does life exist outside the organism?



Claude Bernard (1878) : The physiological systems of an organism can be maintained in a living system after the organism's death.

The period of the precursors (1885-1900) :

Wilhem Roux, chicken embryo marrow; **Prof Ross Harrison** frog neuroblast culture in lymph medium; **Pr Ringer** Ca^{2+} in the extracellular environment is essential for cardiac contractions

The development of tissue culture (from 1902):

Alexis Carrel : Improvement of tissue harvesting techniques, development of aseptic rules, study of nutritional requirements. Development of the first culture vessel (flask).

Cell Culture

Pr Roger Jean Gautheret 1937 : totipotency

Moscona en 1962: introduction of tissue trypsinization.

« Modern » Cell Culture

Pr Monique Adolphe (human cell culture) - start of cell therapy



A- Mechanisms involved in weight gain and adipose tissue damage mediated by new antiretroviral drugs in people living with HIV

STUDY MODELS

- 1. *IN VIVO* studies, on an entire organism**

A- Mechanisms involved in weight gain and adipose tissue damage mediated by new antiretroviral drugs in people living with HIV

IN VIVO cohorte ObeVIH

Obese infected and treated Patients

Pr V Pourcher



Obese Patients
non infected

Infected and treated without INSTI
Infected and treated with INSTI



- ✓ Metabolic monitoring of patients
- ✓ Evaluation of AT distribution DEXA
- ✓ Insulin sensitivity assessment
- ✓ Assessing the impact of bariatric surgery

A- Mechanisms involved in weight gain and adipose tissue damage mediated by new antiretroviral drugs in people living with HIV

STUDY MODELS

1. *IN VIVO* studies, on an entire organism

2. *EX VIVO* studies: on an entire organ/tissue

A COMPLEX cell model



2. *Ex vivo* studies,
On tissue

A complex cell model



Organ or tissue culture :
maintaining integrity

3. *In vitro* studies



Primary culture cells :
dissociation and survival

Cell lines :
limited or
unlimited number of
divisions

A- Mechanisms involved in weight gain and adipose tissue damage mediated by new antiretroviral drugs in people living with HIV

EX VIVO

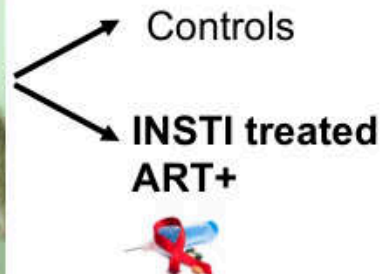
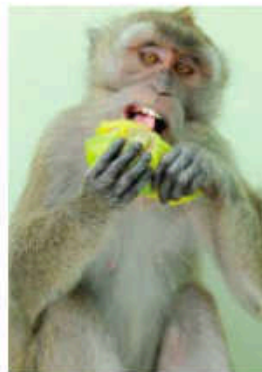
Cynomolgus macaque (*Macaca fascicularis*)

Primate Research Institute
Dr C Bourgeois/Pr O Lambotte

Infected and treated obese patients

Cohorte ObéVIH Pr V
Pourcher

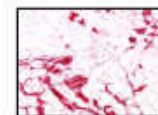
CONTROL group!!!!!!



Obese patients
without infection

Infected and treated without INI
Infected and treated with INI

SCAT and VAT



VAT: Visceral Adipose Tissue
SCAT: Subcutaneous Adipose Tissue

Gorwood J, et al., AIDS 2019; CID 2020; Cells 2020

Ngono Ayissi K. et al., Cells 2022



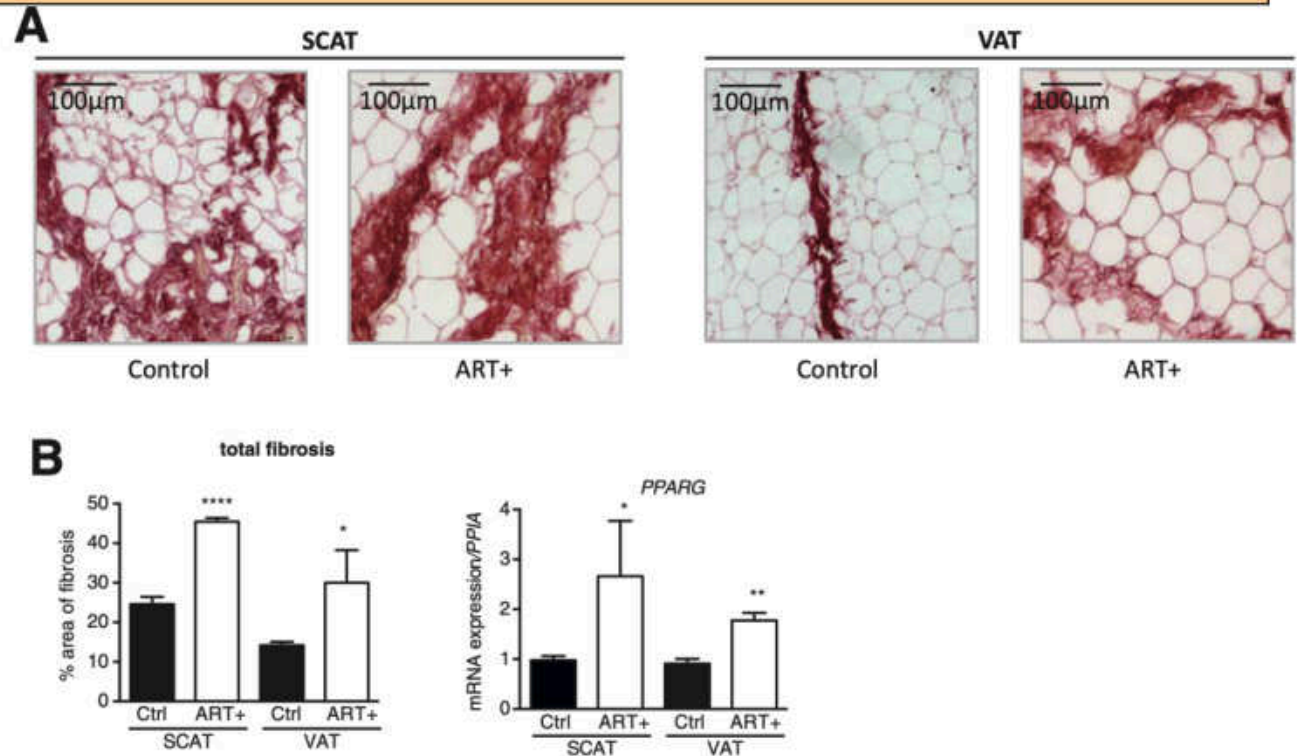
Claire Lagathu MCU
Véronique Béréziat PU



Jennifer Gorwood Master BIP, Ph.D.
Sidaction
Postdoctoral fellowship in Sweden
and in CRSA

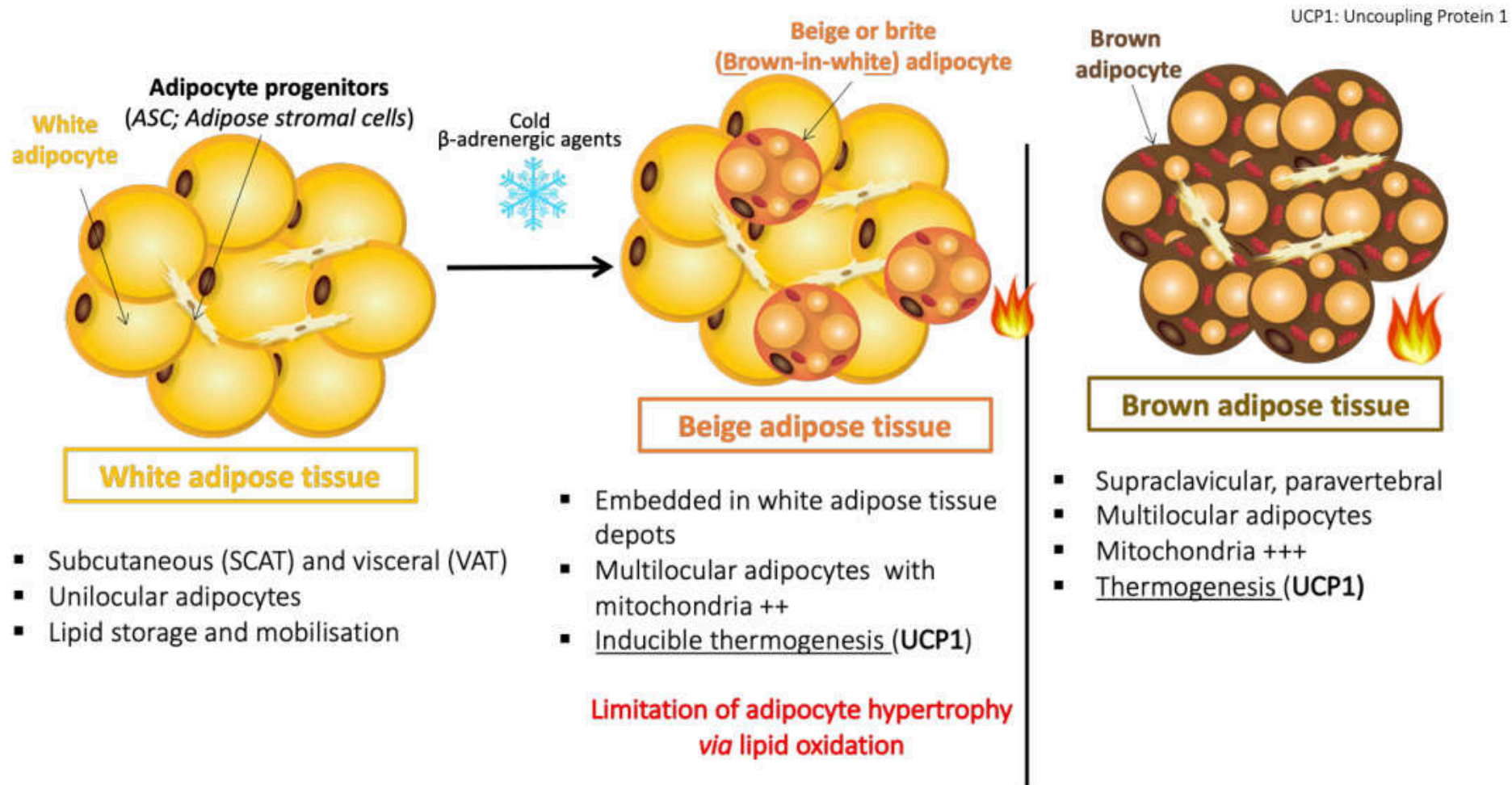
Gorwood J. et al., AIDS 2019.

VAT: Visceral Adipose Tissue
SCAT: Subcutaneous Adipose Tissue
ART + : Antiretroviral therapy +



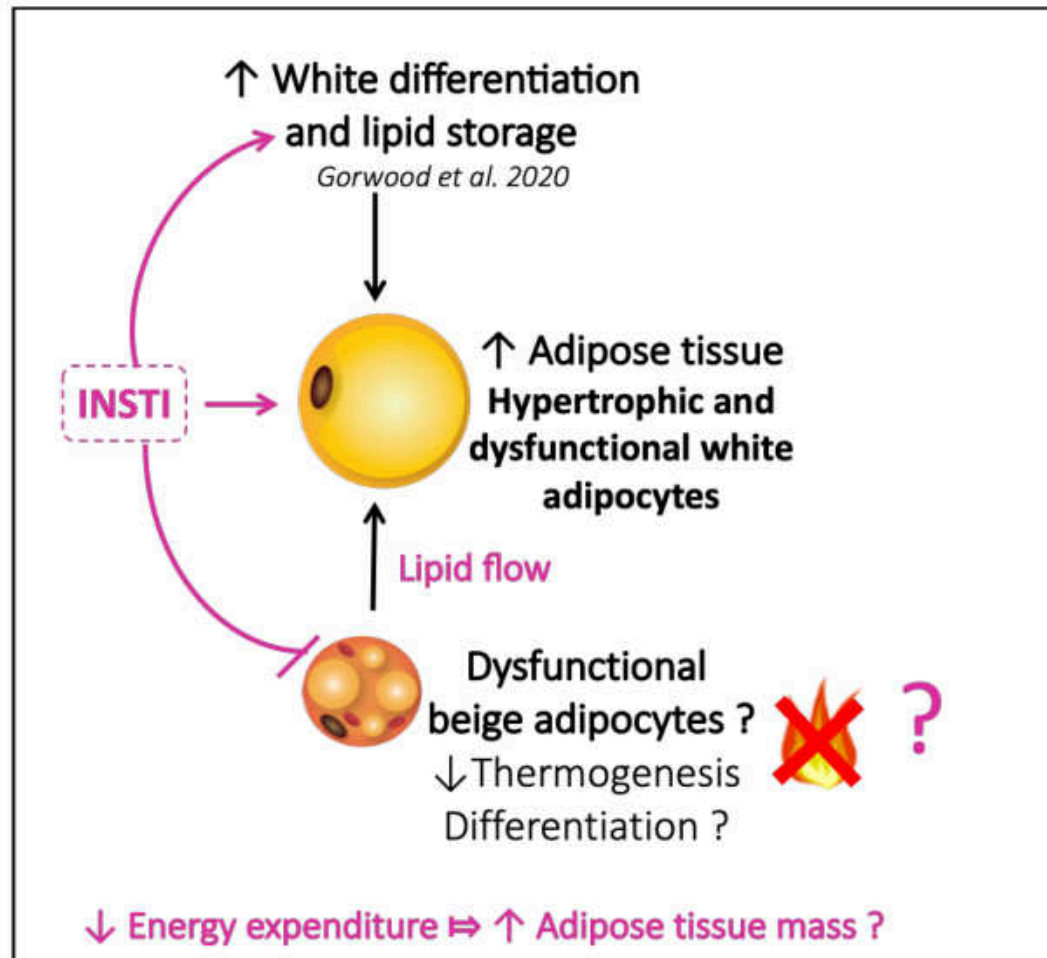
➤ SCAT and VAT From INSTI-Treated Macaques (ART+) Display Elevated Fibrosis, Large Adipocytes, and High Levels of Adipogenic Markers

A- Mechanisms involved in weight gain and adipose tissue damage mediated by new antiretroviral drugs in people living with HIV



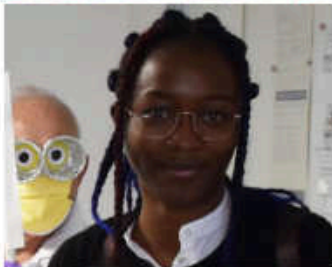
A- Mechanisms involved in weight gain and adipose tissue damage mediated by new antiretroviral drugs in people living with HIV

Question asked: Do INIs inhibit the ability to beige?



A- Mechanisms involved in weight gain and adipose tissue damage mediated by new antiretroviral drugs in people living with HIV

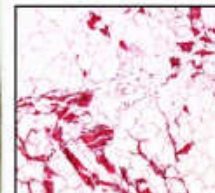
Claire Lagathu MCU
Véronique Béréziat PU



Kenza Ngono Ayissi,
doctorat Sidaction

SIVART Cohort

Collaboration: O. Lambotte, C. Bourgeois, R. Le Grand (UMRS1184/CEA)
SIV-infected macaques receiving TDF/FTC/**DTG** (n=9 ; ARV
duration 2.32 ± 0.04 years ; Age: 9.8 ± 0.2 years ; 8.9 ± 0.3 kg)
vs control macaques (n=6 ; Age: 7.9 ± 1.1 years ; 9.8 ± 1.4 kg)

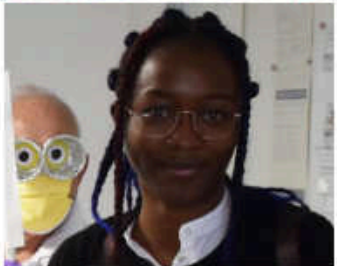


SCAT
VAT

VAT: Visceral Adipose Tissue
SCAT: Subcutaneous Adipose Tissue
SIVART: SIV + Antiretroviral therapy +
ARV: antiretroviral

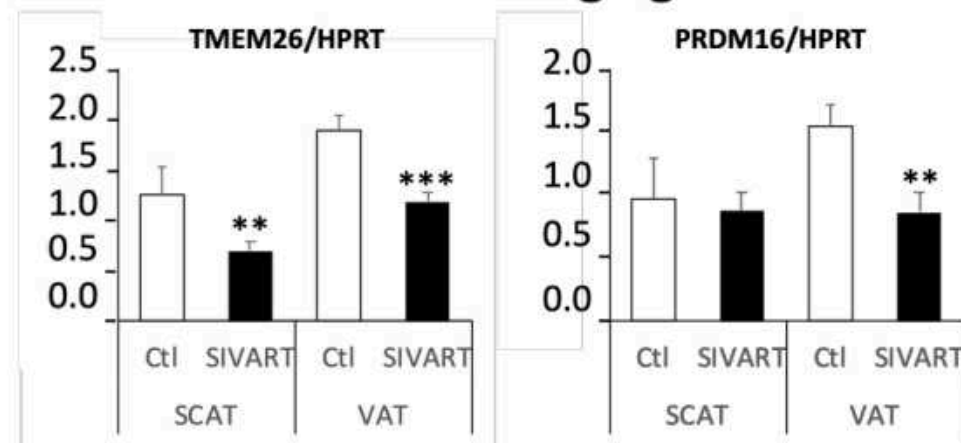
A- Mechanisms involved in weight gain and adipose tissue damage mediated by new antiretroviral drugs in people living with HIV

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doctorat Sidaction

mRNA levels of beiging markers



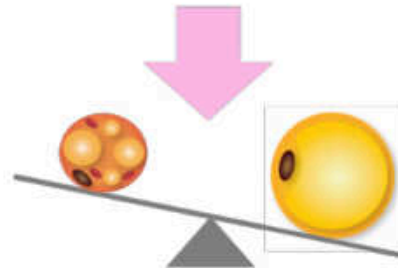
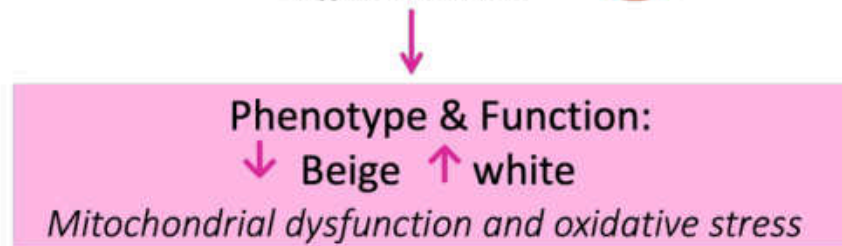
- SCAT and VAT From SIVART Macaques (SIV+ INSTI+) Display decreased beiging markers expression TMEM 26 and PRDM16

VAT: Visceral Adipose Tissue
SCAT: Subcutaneous Adipose Tissue
SIVART: SIV + Antiretroviral therapy +
ARV: antiretroviral

A- Mechanisms involved in weight gain and adipose tissue damage mediated by new antiretroviral drugs in people living with HIV



ASC: Adipose stromal cells
DTG: Dolutegravir
BIC: Bictegravir



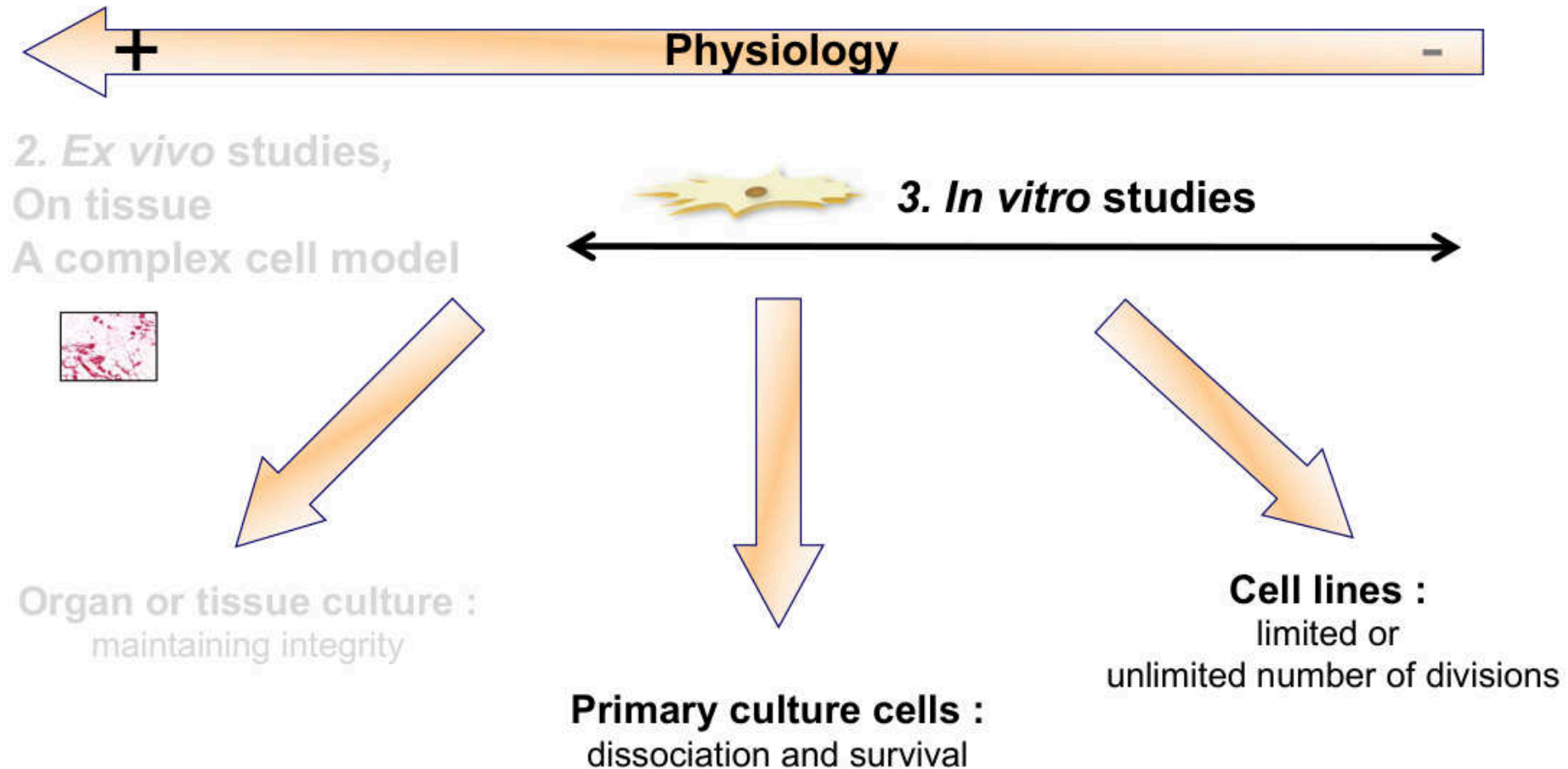
Limit of the study?

Accumulation of hypertrophic and dysfunctional white adipocytes

↑Adipose tissue mass and weight gain

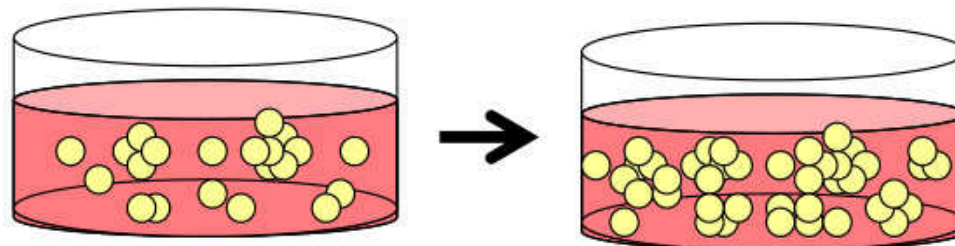
III- *In vitro* models definition

STUDY MODELS

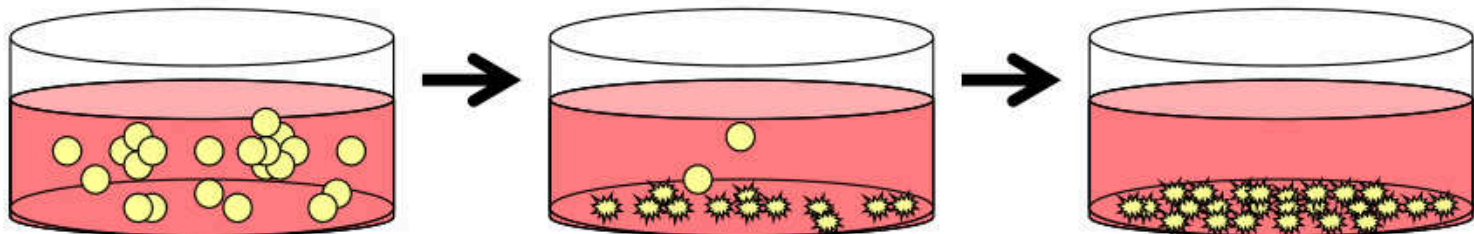


Suspended or adherent cells?

Suspension cell culture (lymphocytes, bone marrow, certain types of tumors such as small round cell tumors, etc.) : the cells float in the medium and proliferate in suspension.

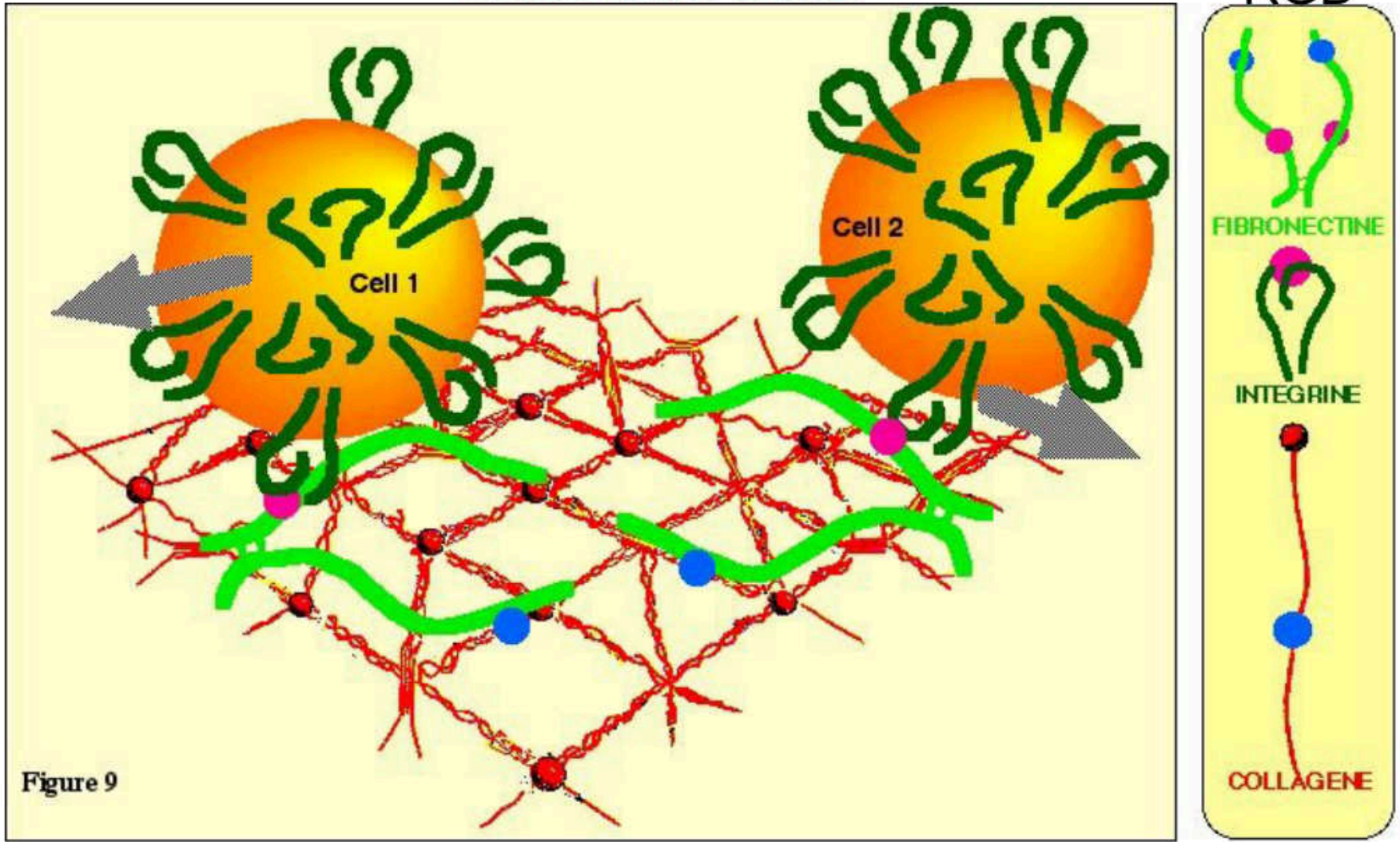


Substrate culture : adhesion of cells to the wall at the bottom of the flask or culture dish



III-1 Cell biology : cell adhesion

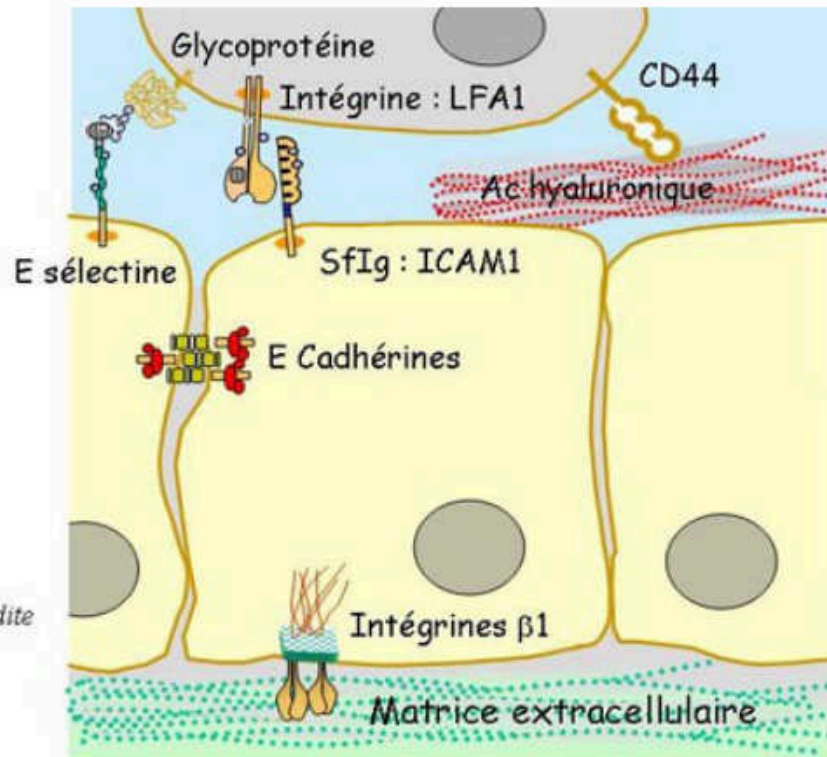
ExtraCellular Matrix



Signaling downstream of integrins :

Focal Adhesion Kinase (FAK), Src, Growth Factor Receptors

III-1 Cell biology: cell adhesion

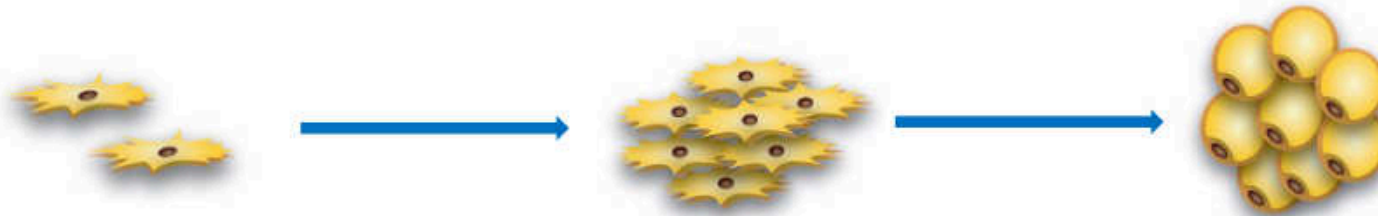


CAM	Super Famille des Ig
	Cadhérines
	Sélectines
CAM / SAM	Intégrines

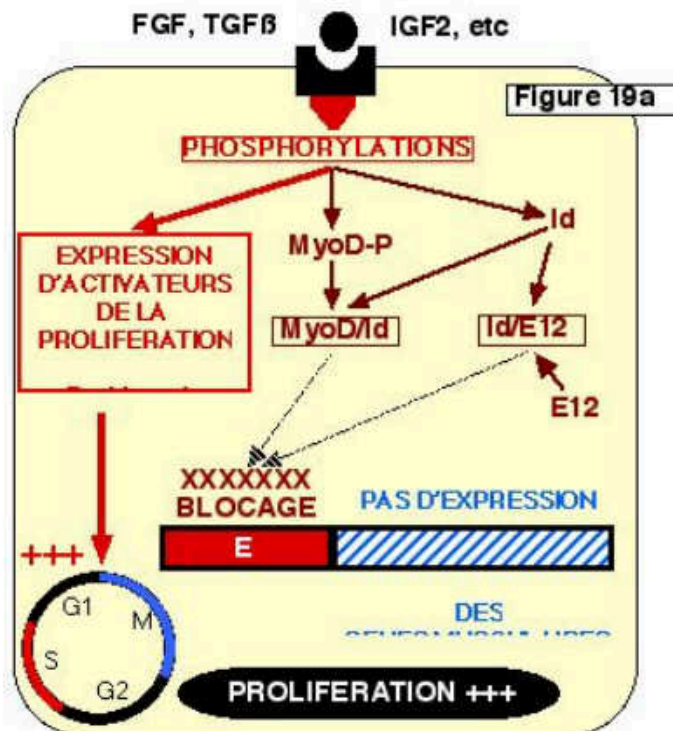
CAM: Cell Adhesion Molecule
SAM: Substrate Adhesion Molecule

11

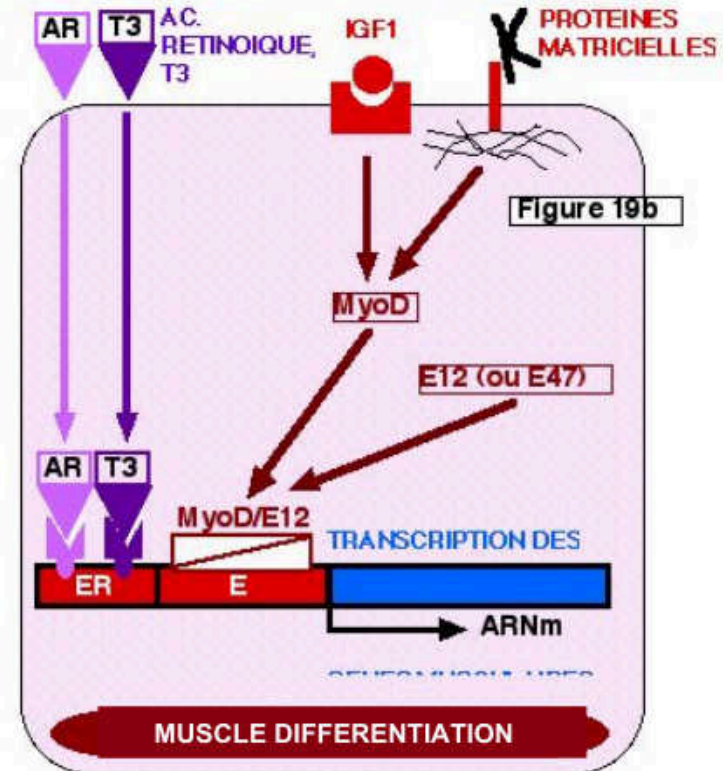
III-2 Cell biology: switch proliferation versus differentiation



Growth Factors



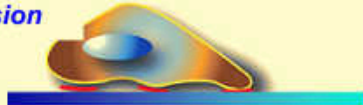
Differentiation Factors



III-2 Cell biology: migration

Cell migration

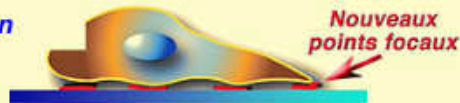
1- Adhesion



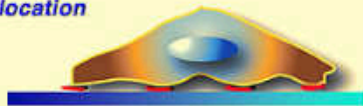
2-Extension



3- New Adhesion



4-Translocation

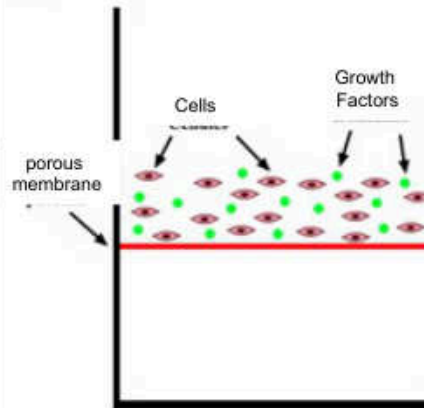


5- detachment

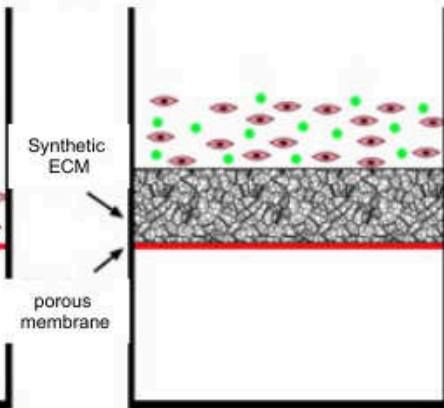


Direction of movement

A

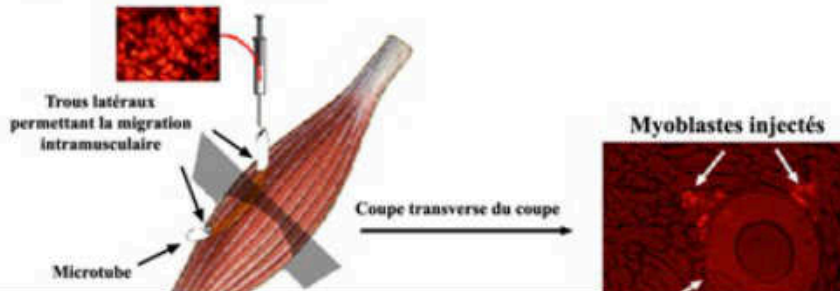


B

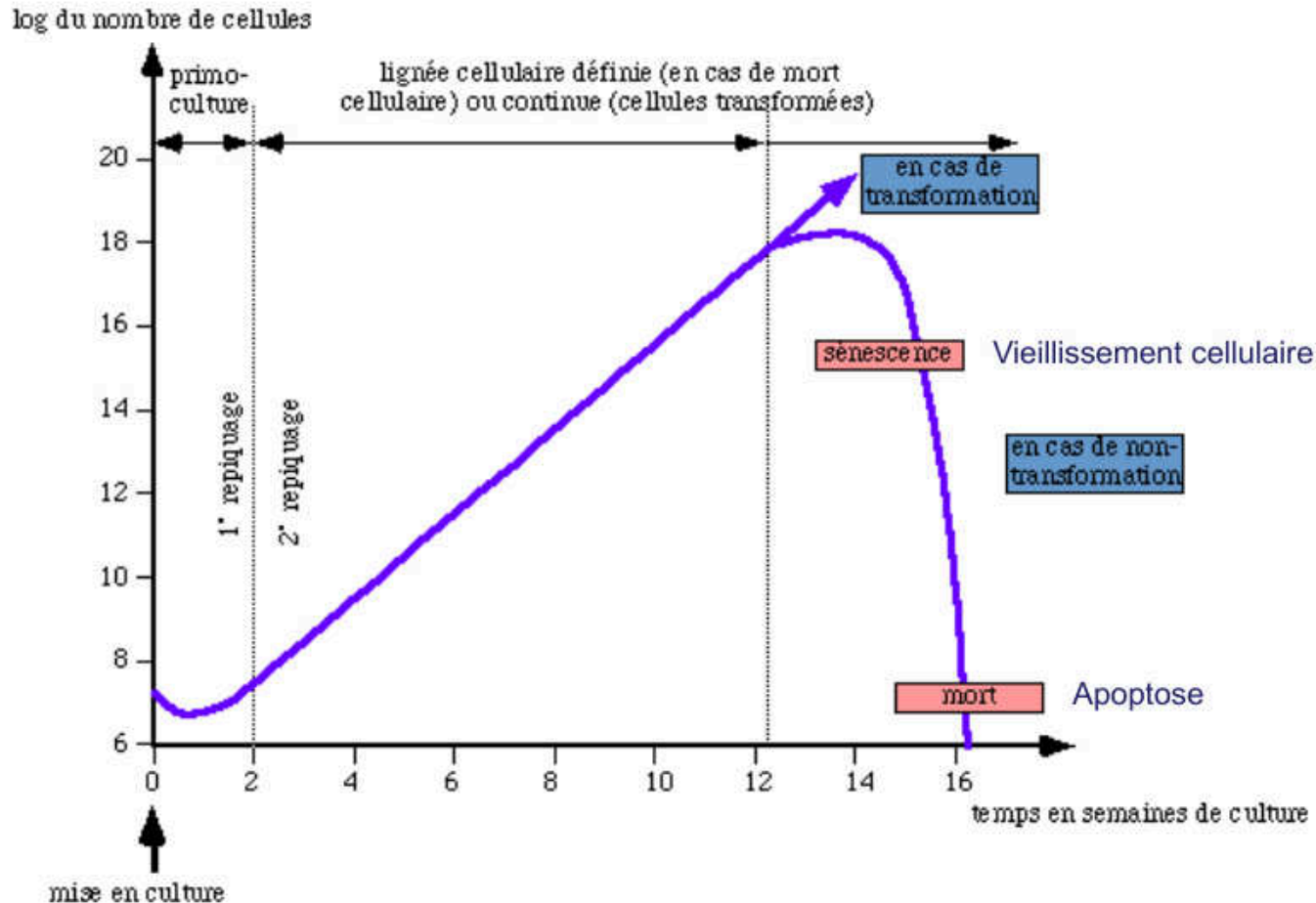


C

Myoblastes marqués au PKH-26



Establishment of a cell line



Immortalized cells :

- preserved contact inhibition
- gives the cell unlimited capacity for proliferation
- no phenotypic transformation

single-layer cell culture

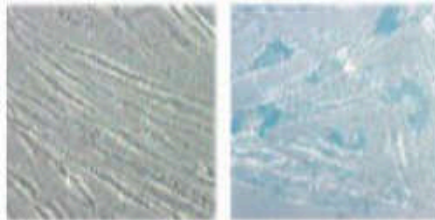
Transformed cells :

- loss of contact inhibition
- loss of dependence on the support

multilayer cell culture

III-3 Cell biology: cellular senescence

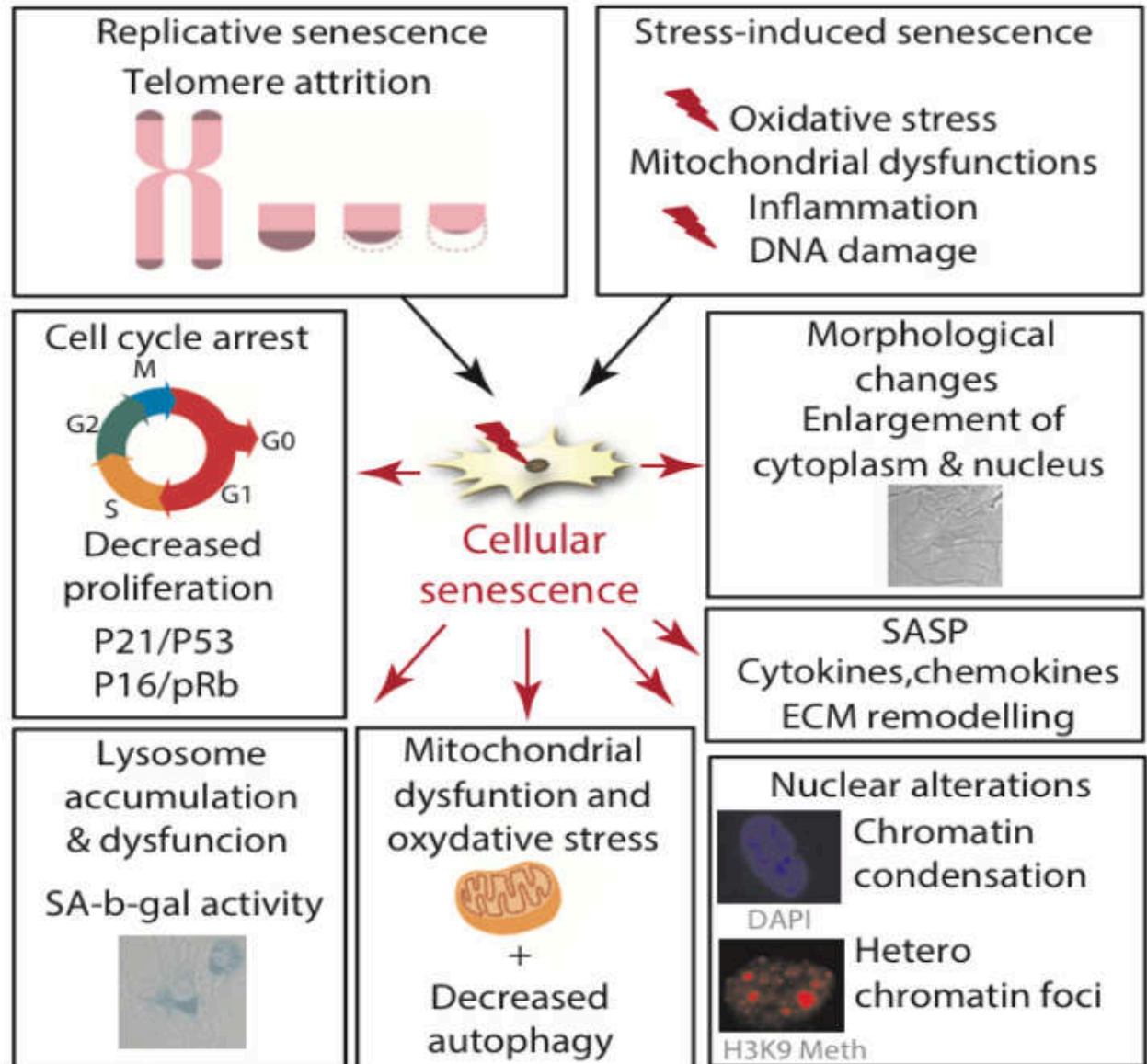
Senescent associated β -galactosidase activity



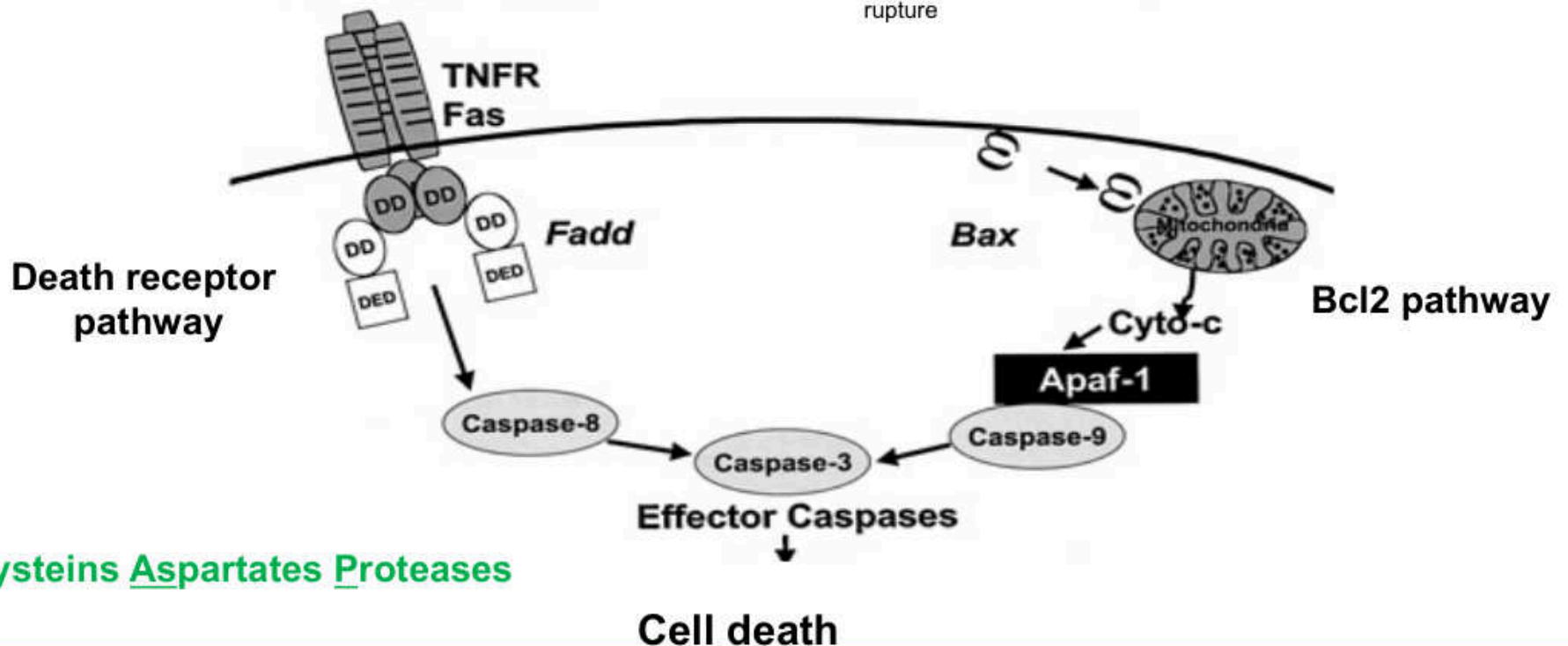
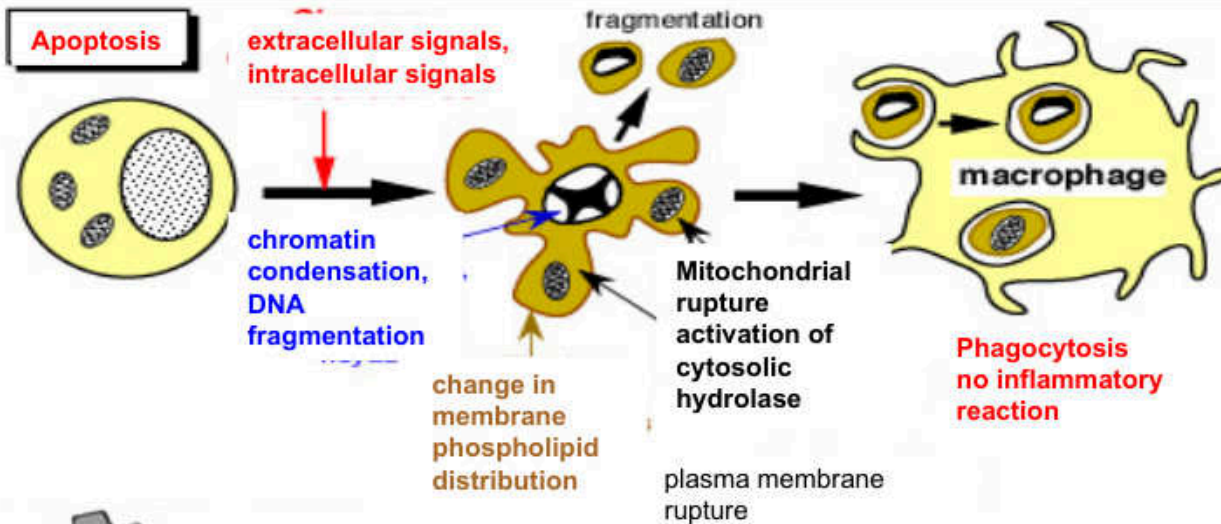
Contrôle

R482W

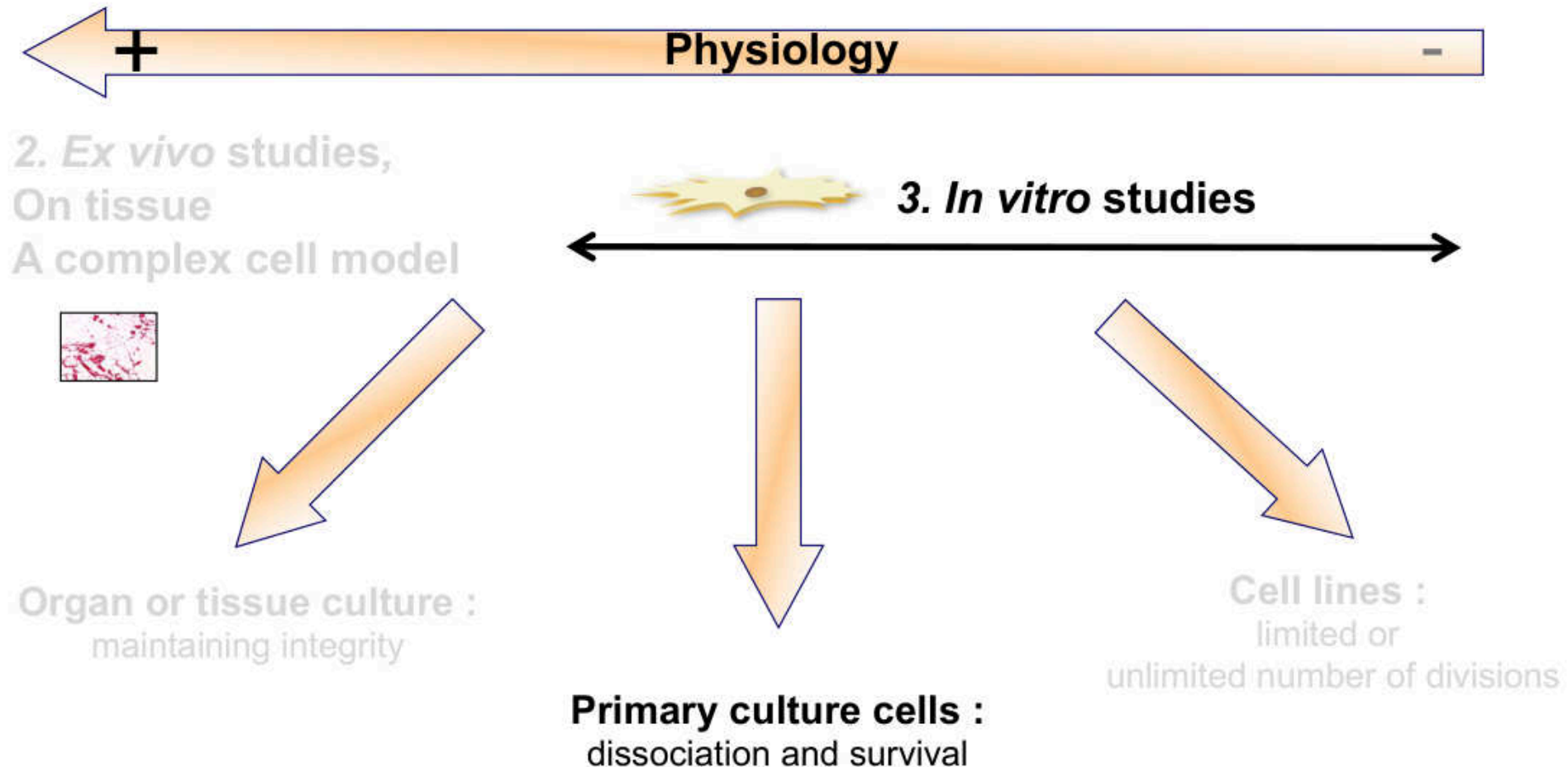
Caron, CDD, 2007



III-4 Cell biology: apoptosis



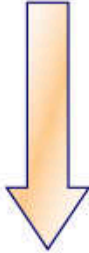
STUDY MODELS: primary culture cells



IV- *In vitro* models- primary culture

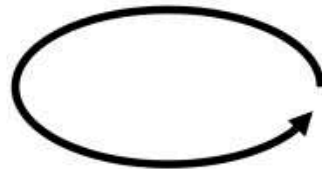
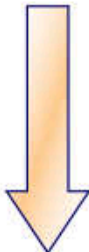
Primary culture: isolation of an adherent cell type within tissue

- **Mecanical** and/or **enzymatic** dissociation
of fragment tissue:



*Scissors, syringe
trypsin, collagenase, dispase*

- Cell Filtration and Separation

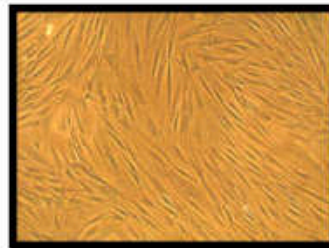
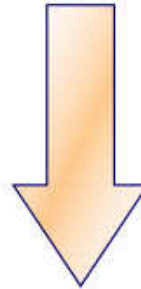
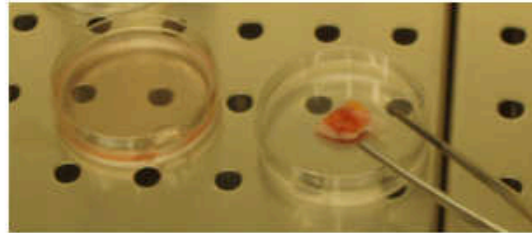


*Centrifugation, surface proteins,
nutrients*

-dhesion of isolated cells in a specific
culture medium

Primary culture: isolation of an adherent cell type within tissue

Special case of skin
fibroblasts



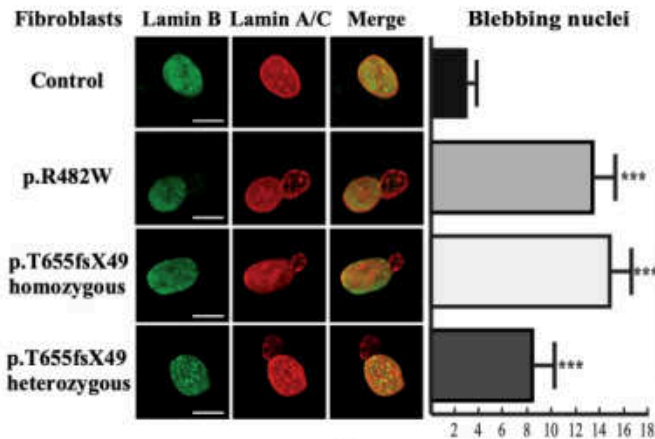
- **Migration** from an explant
- **Limited proliferative capacity**

AIM Characterization of a new mutation in the *LMNA* gene responsible for lipodystrophy

Corinne Vigouroux
PUPH

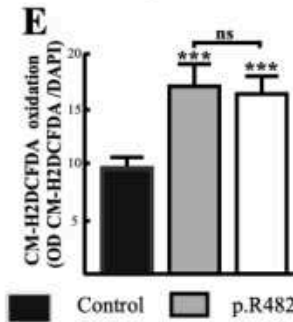


Caroline Le Dour
Master BIP, doctorat
Postdoctoral fellowship in
USA



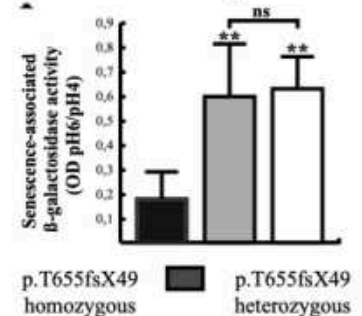
Oxidative Stress

Mesure of reactive oxygen species production using the DCFDA probe



Senescence

Mesure of senescent associated β -galactosidase activity



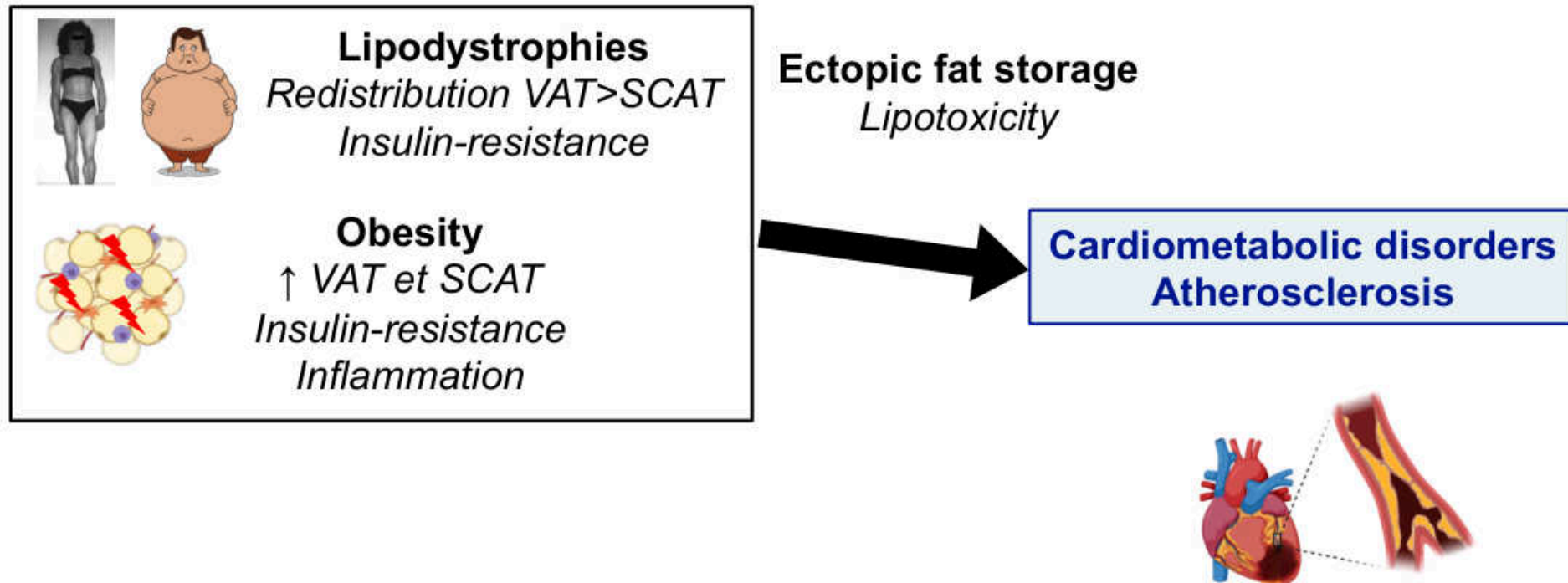
RESULTS

Fibroblasts from lipodystrophic patients carrying the p.T655fsX49 mutation exhibit

- an increase in the percentage of blebbing nuclei
- an increase in the percentage of blebbing nuclei
- oxidative stress and a cellular senescence

B- Impact of adipose tissue aging on the development of atherosclerosis

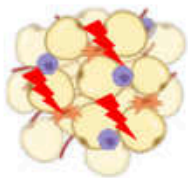
1. Context





Lipodystrophies

Redistribution VAT>SCAT
Insulin-resistance



Obesity

$\uparrow$ VAT et SCAT
Insulin-resistance
Inflammation

Ectopic fat storage
Lipotoxicity

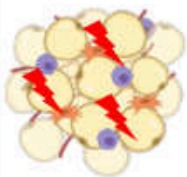


Cardiometabolic disorders
Atherosclerosis

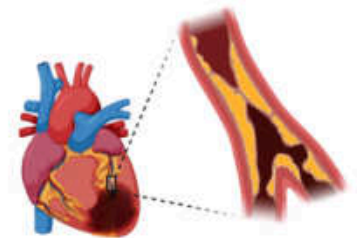


Aging

Redistribution VAT>SCAT
Fibrosis & inflammation

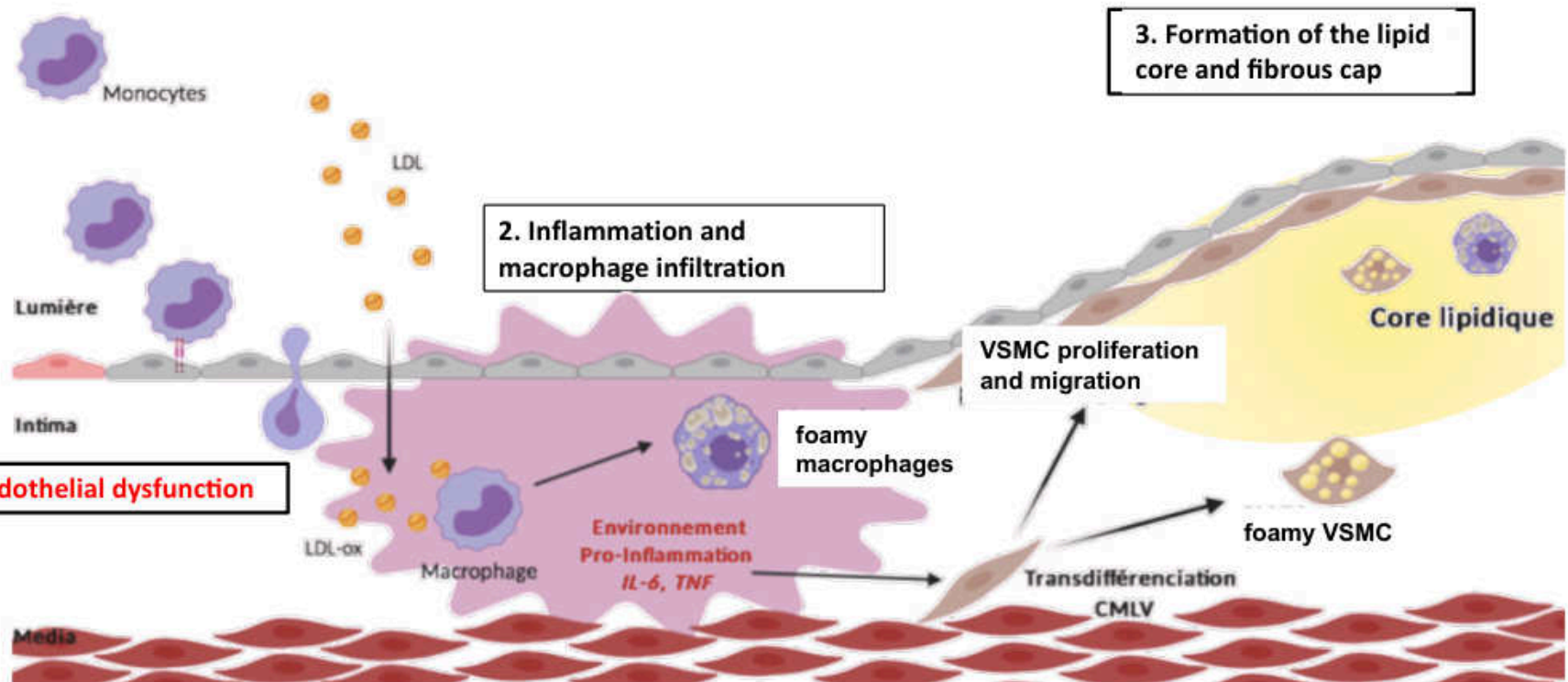


Dysfunctional ASC



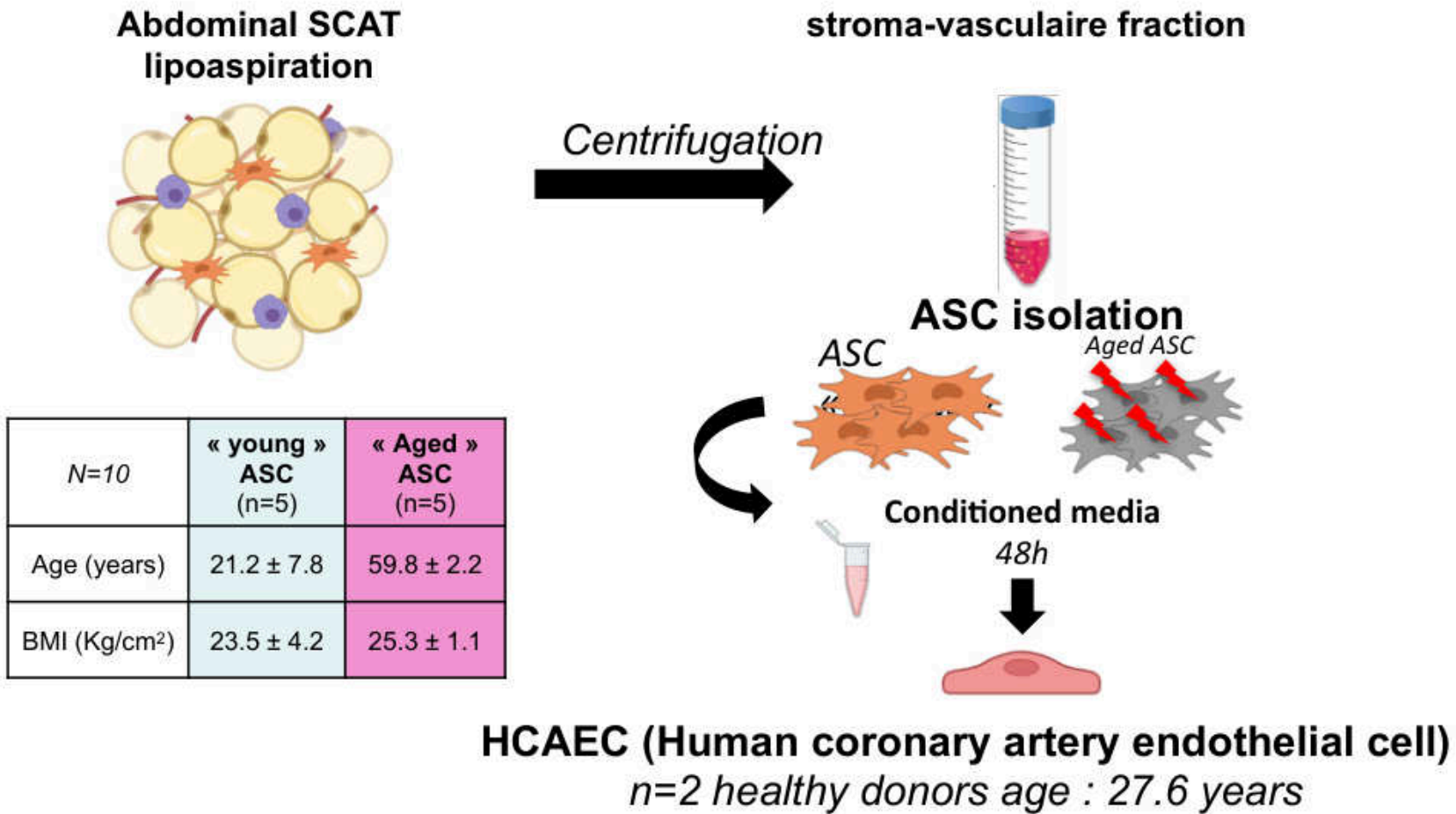
**2. Do ASC dysfunctions
observed during aging
promote atherosclerosis??**

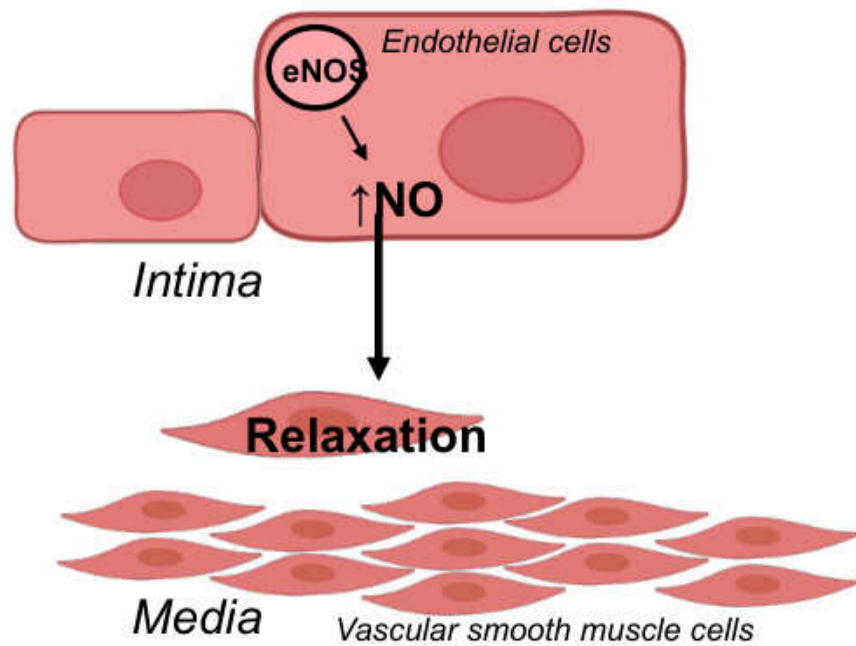
Atherogenesis



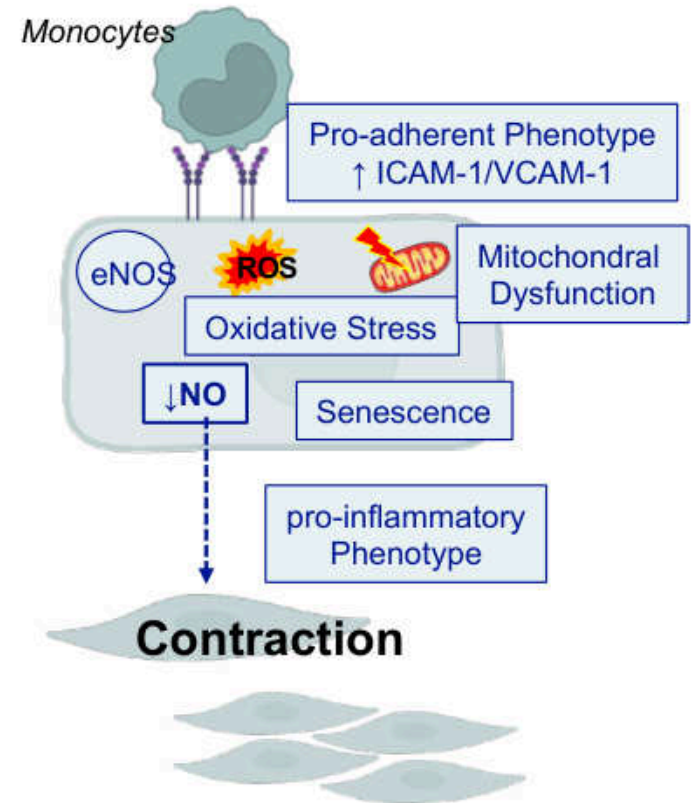
LDL ox : low density lipoprotein oxydée
 VSMC : Vascular Smooth Muscle Cells

Inter organ communication





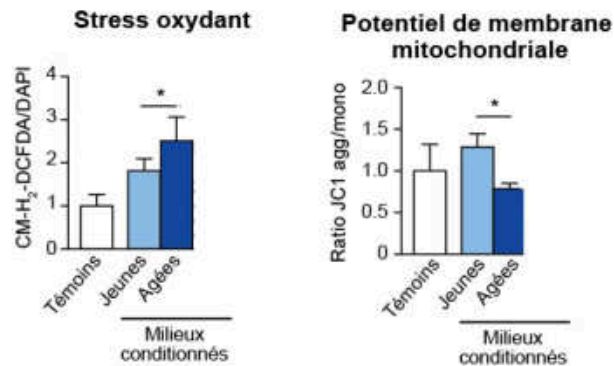
1. Endothelial dysfunction



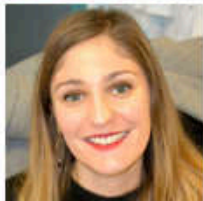
RESULTS

Oxidative stress and mitochondrial dysfunction

ROS Production



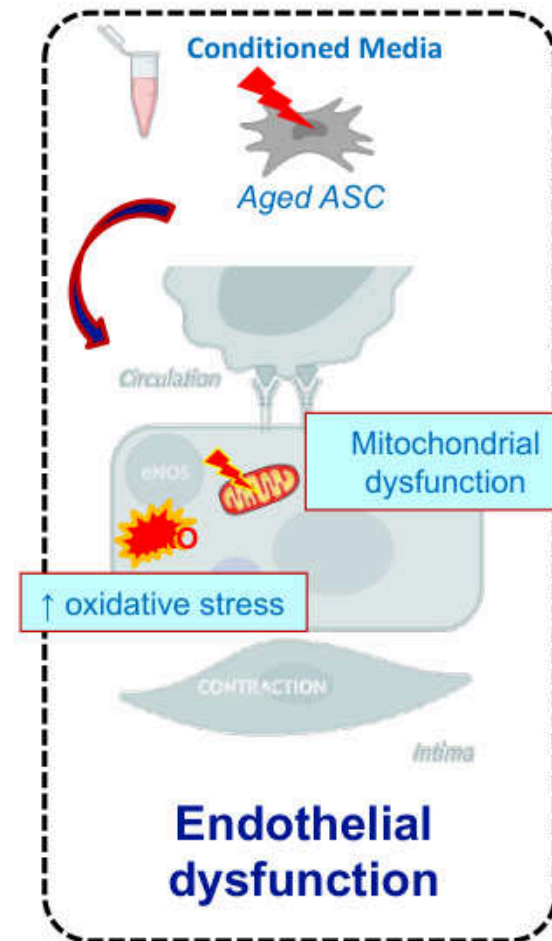
Claire Lagathu MCU
Véronique Béréziat PU



Laura Le Pelletier
Master BIP, PhD
Postdoctoral fellowship in Toulouse

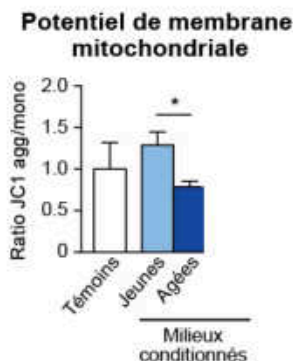
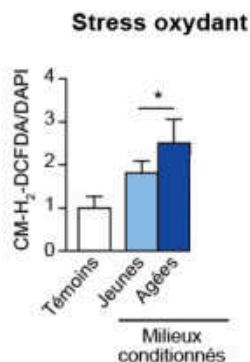
- ☐ HCAEC sans MC
- ☒ HCAEC + MC ASC jeunes
- ☒ HCAEC + MC ASC âgées
- ☒ HCAEC + MC ASC âgées traitées pravastatine

IV- *In vitro* models- primary culture

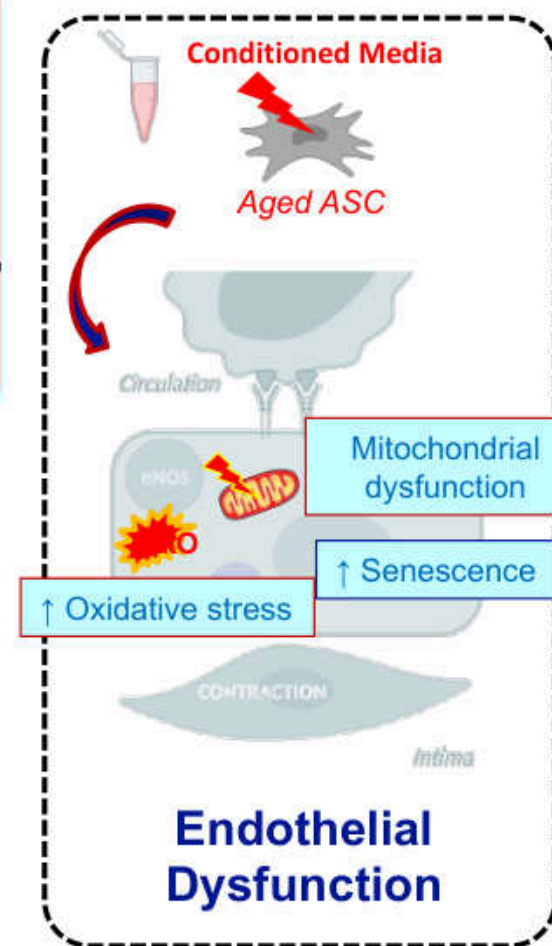
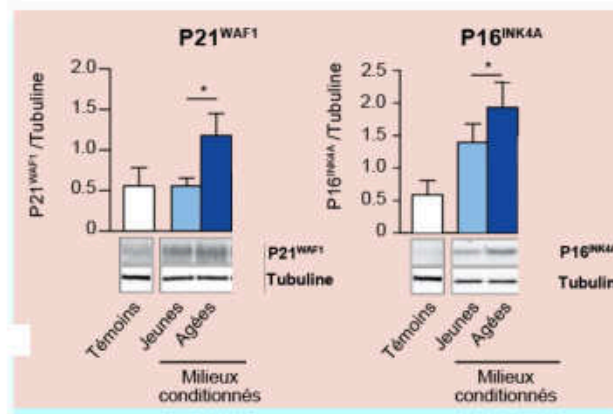


Oxidative Stress and mitochondrial dysfunction

ROS Production



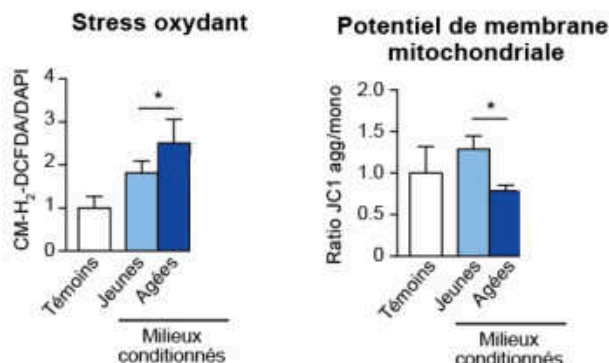
Senescence



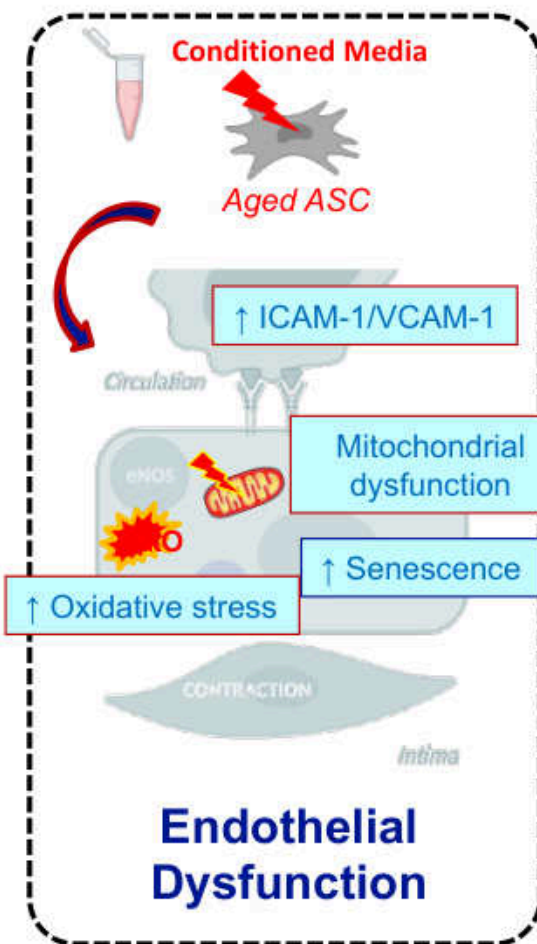
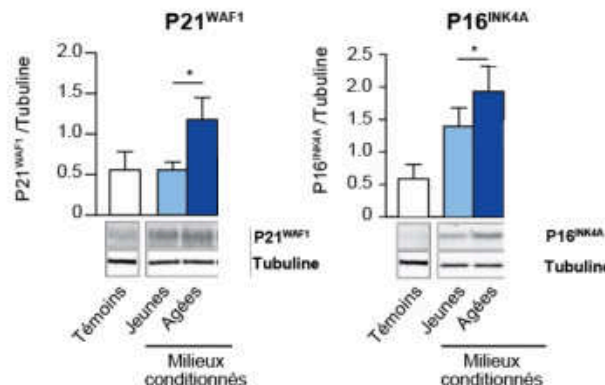
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- ▨ HCAEC + MC ASC âgées traitées pravastatine

Oxidative Stress and mitochondrial dysfunction

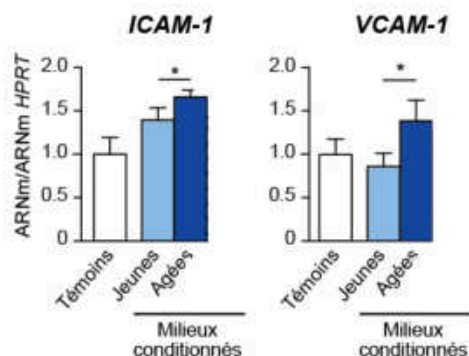
ROS Production



Senescence



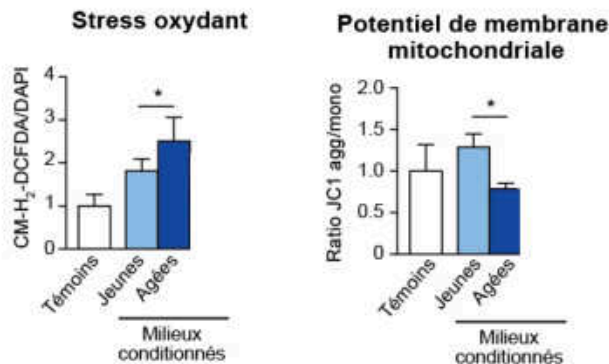
Pro-adherent phenotype



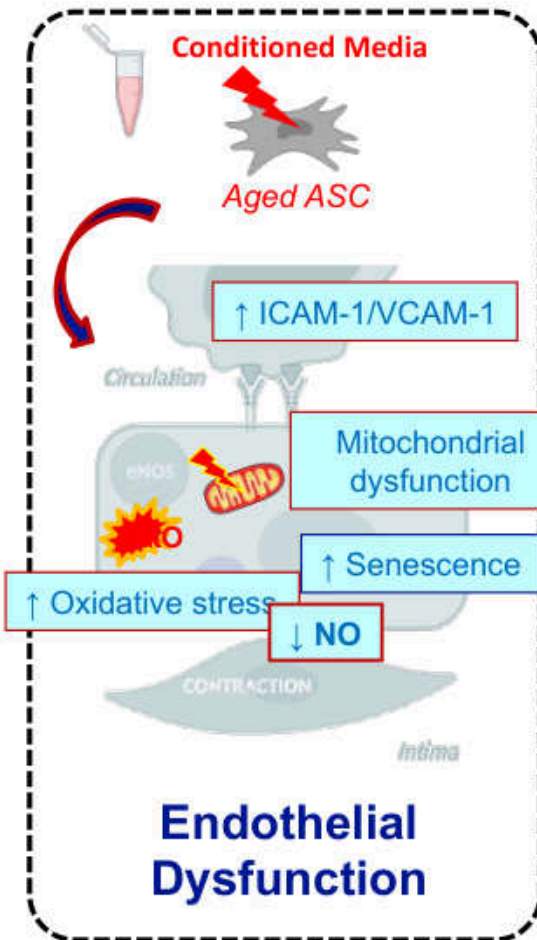
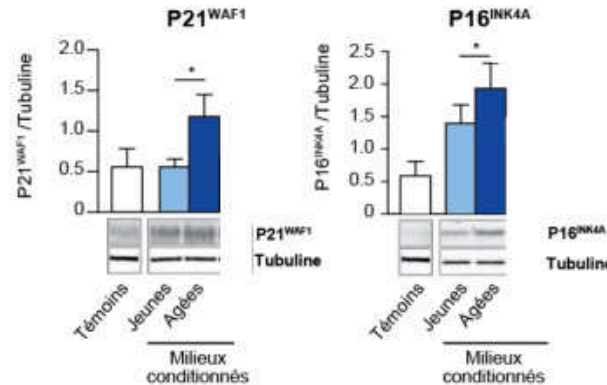
- HCAEC sans MC
- HCAEC + MC ASC jeunes
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- ▨ HCAEC + MC ASC âgées traitées pravastatine

Oxidative Stress and mitochondrial dysfunction

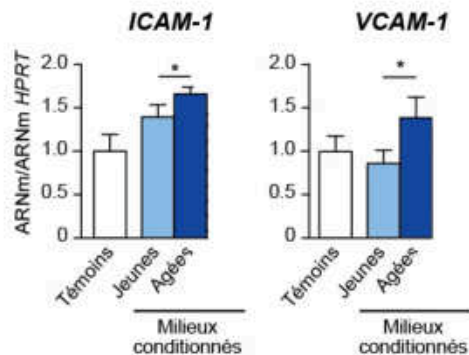
ROS Production



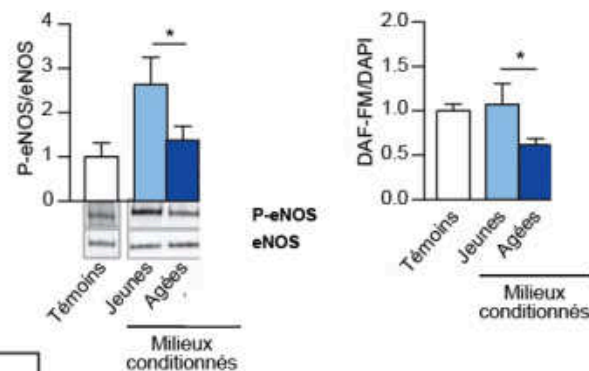
Senescence



Pro-adherent phenotype

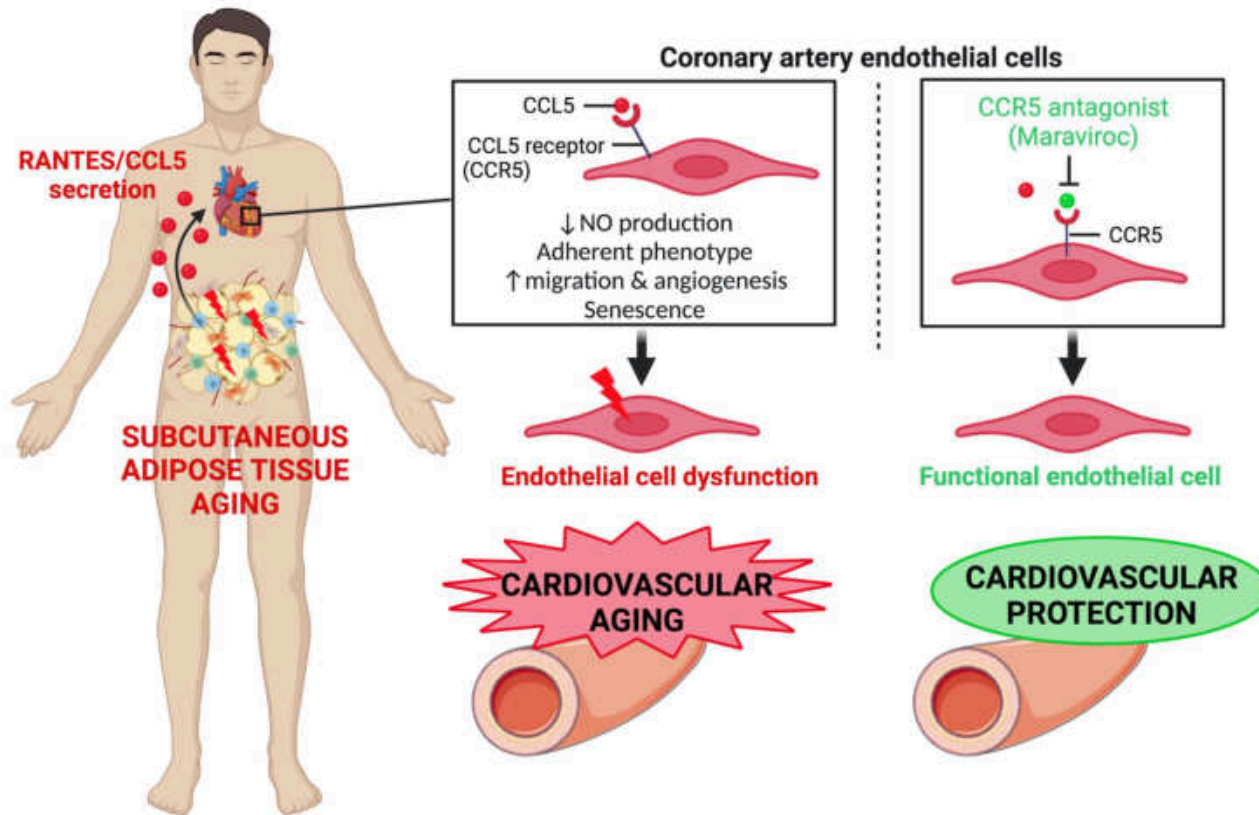


Decrease in NO production



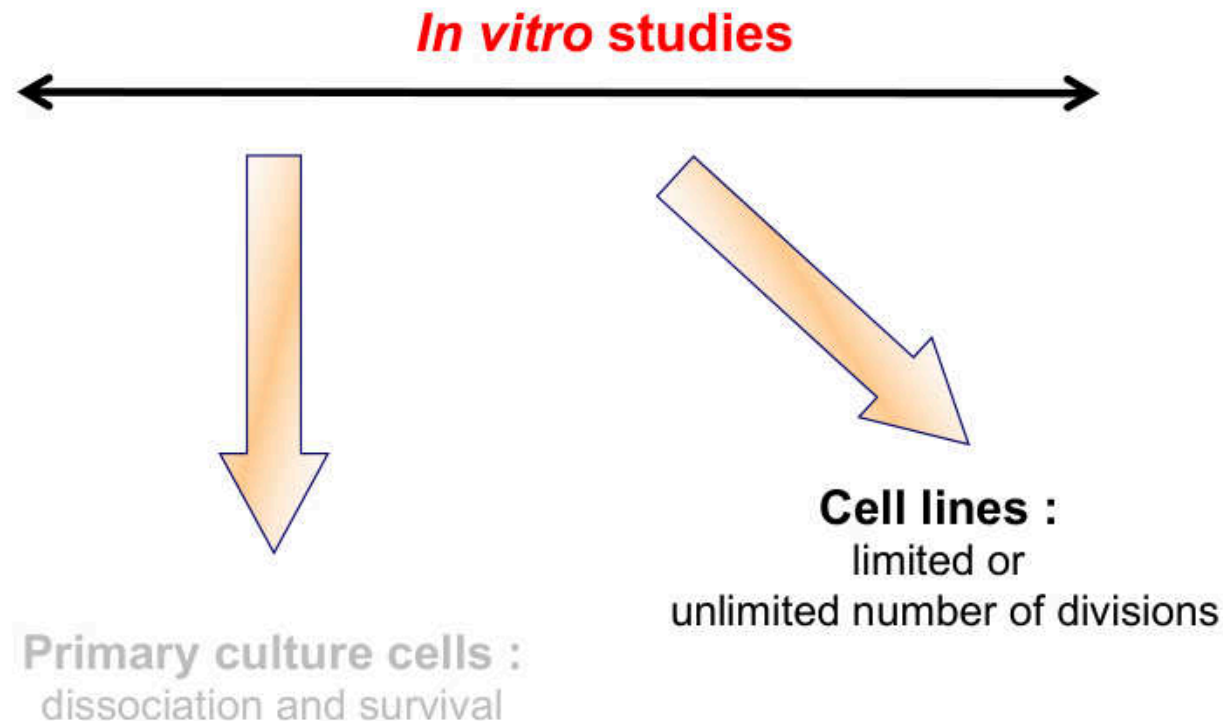
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Inter organ communication



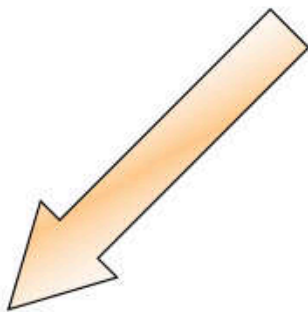
V- *In vitro* models- cell lines culture

Histological culture is primarily aimed at **studying the cellular units** that make up the various tissues of organ, **the cell proliferation** without maintaining their tissue organization or specialized function.

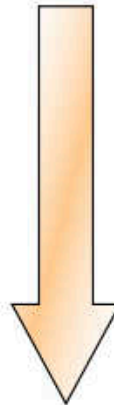


V-1 *In vitro* models- cell lines culture: type of cell lines

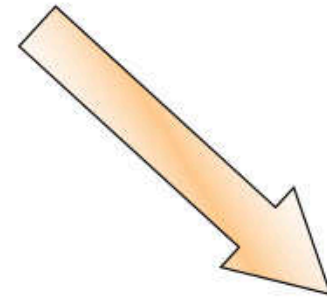
Unlimited proliferative capacity



**Spontaneously
established lineage**



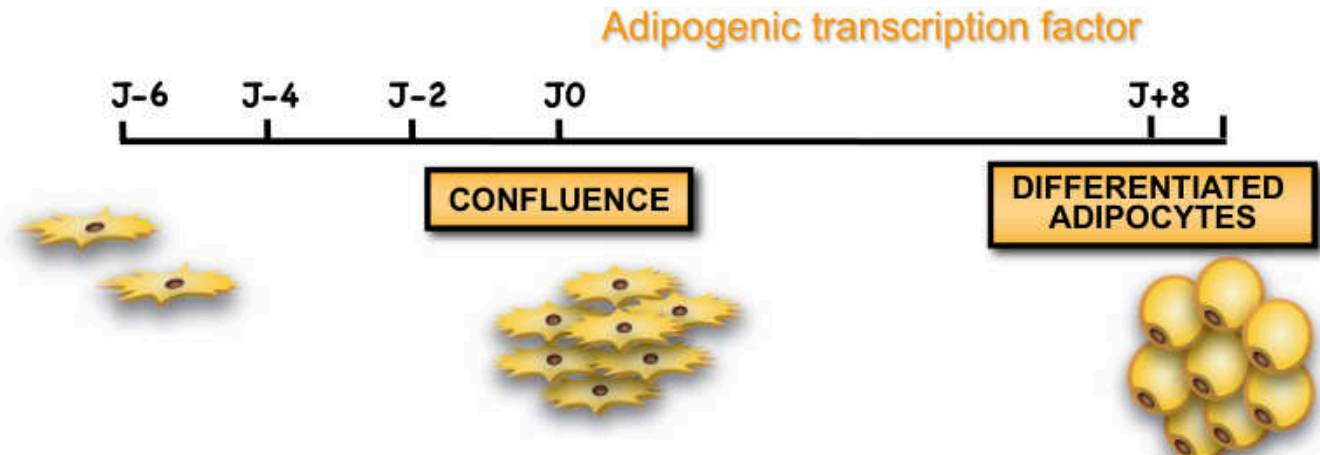
Immortalized lineage :
-cancerous
- infected with a virus



Transformed lines :
genetically modified

V-1 *In vitro* models- cell lines culture: type of cell lines

Spontaneously established lineage



The murine adipocyte cell line 3T3-F442A / 3T3-L1

Immortalized cell lines

Cancer cell lines

Human lineage :

- Hela, resulting from cervical cancer
- Caco-2, resulting from colon cancer
- MCF-7, resulting from breast cancer

Mouse lines

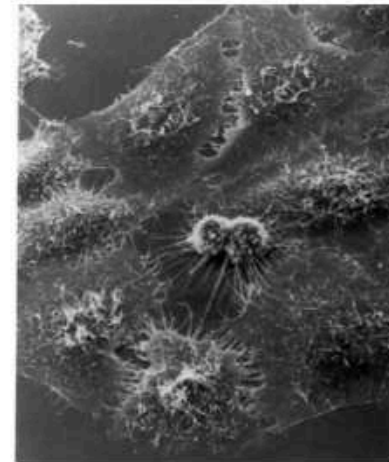
- AtT20, resulting from a corticotrophic pituitary tumor

Rat lines

- AtT20, resulting from a somatotrophic pituitary tumor

Lines immortalized by an oncogene

- cMyc
- AgT SV40
- Adenovirus E1A



V-1 *In vitro* models- cell lines culture: type of cell lines

Immortalized cell lines

Cancer cell lines

Human lineage :

- Hela, resulting from cervical cancer
- Caco-2, resulting from colon cancer
- MCF-7, resulting from breast cancer

Mouse lines

- AtT20, resulting from a corticotrophic pituitary tumor

Rat lines

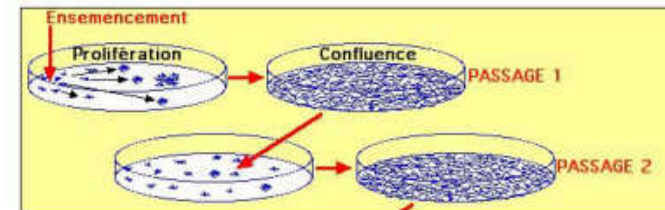
- AtT20, resulting from a somatotrophic pituitary tumor

Lines immortalized by an oncogene

- cMyc
- AgT SV40
- Adénovirus E1A

Contact inhibition preserved
No phenotype transformation

→ Single-layer cell culture



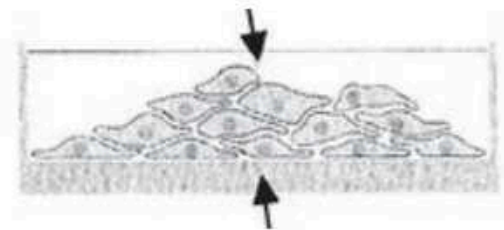
Transformed cell lines

Lines transformed by an oncogene

- Ras
- Polyomavirus middle-T
- Adenovirus E1B

Loss of contact inhibition
Loss of dependence on the medium
for phenotype transformation

➔ Multilayered cell culture



Unlimited proliferative capacity

V-1 *In vitro* models- cell lines culture: type of cell lines

Growth dynamics of a cell culture

Latency phase :

- No cell division
- Depends on cell type, culture medium, density

Exponential phase

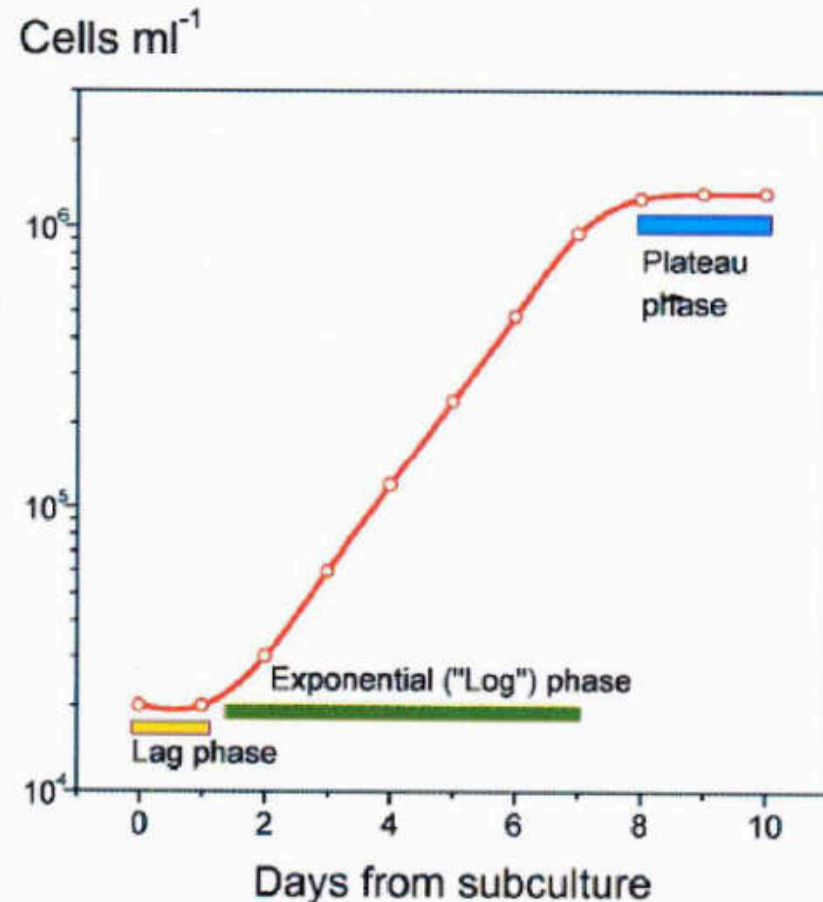
- Cell division phase
- Doubling time according to cell type

Steady state

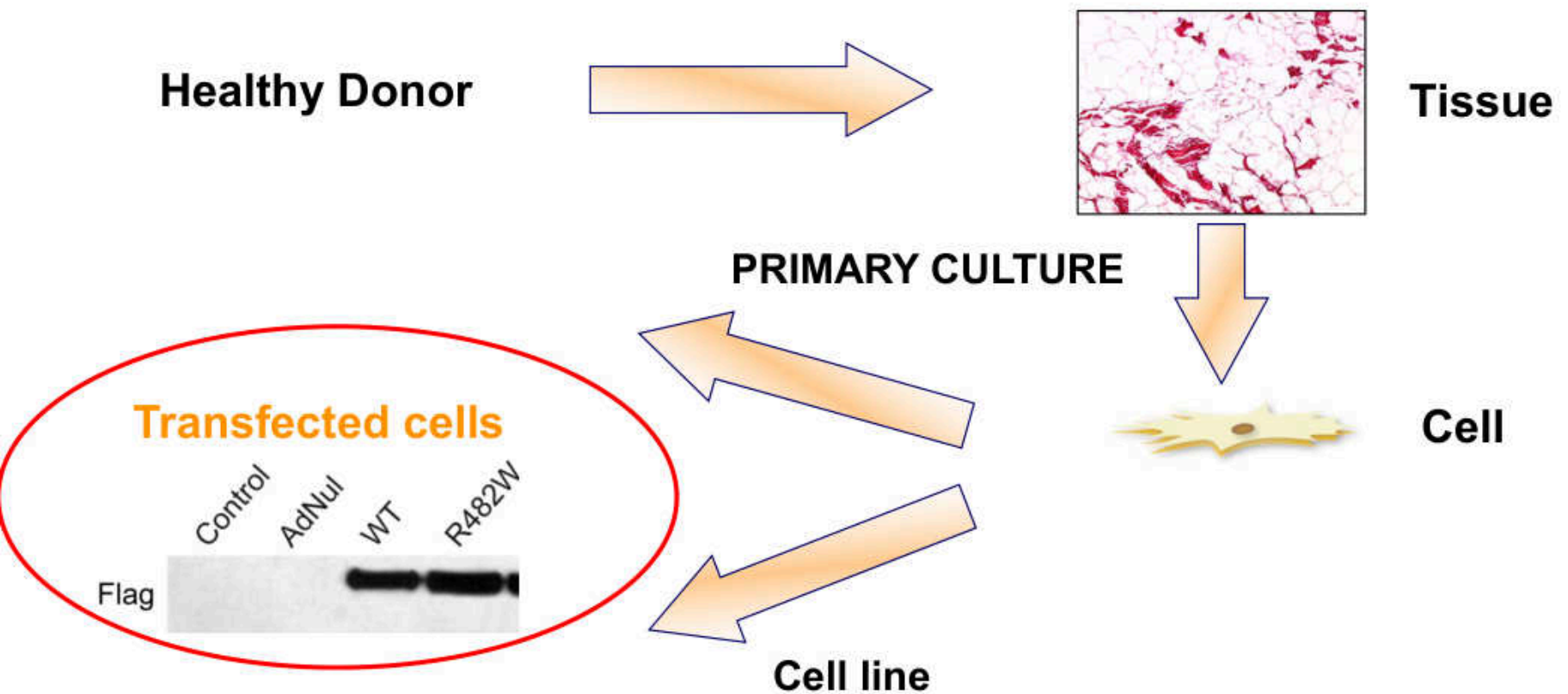
- Depletion of the environment = proliferation shutdown

Continued growth (transplanting)

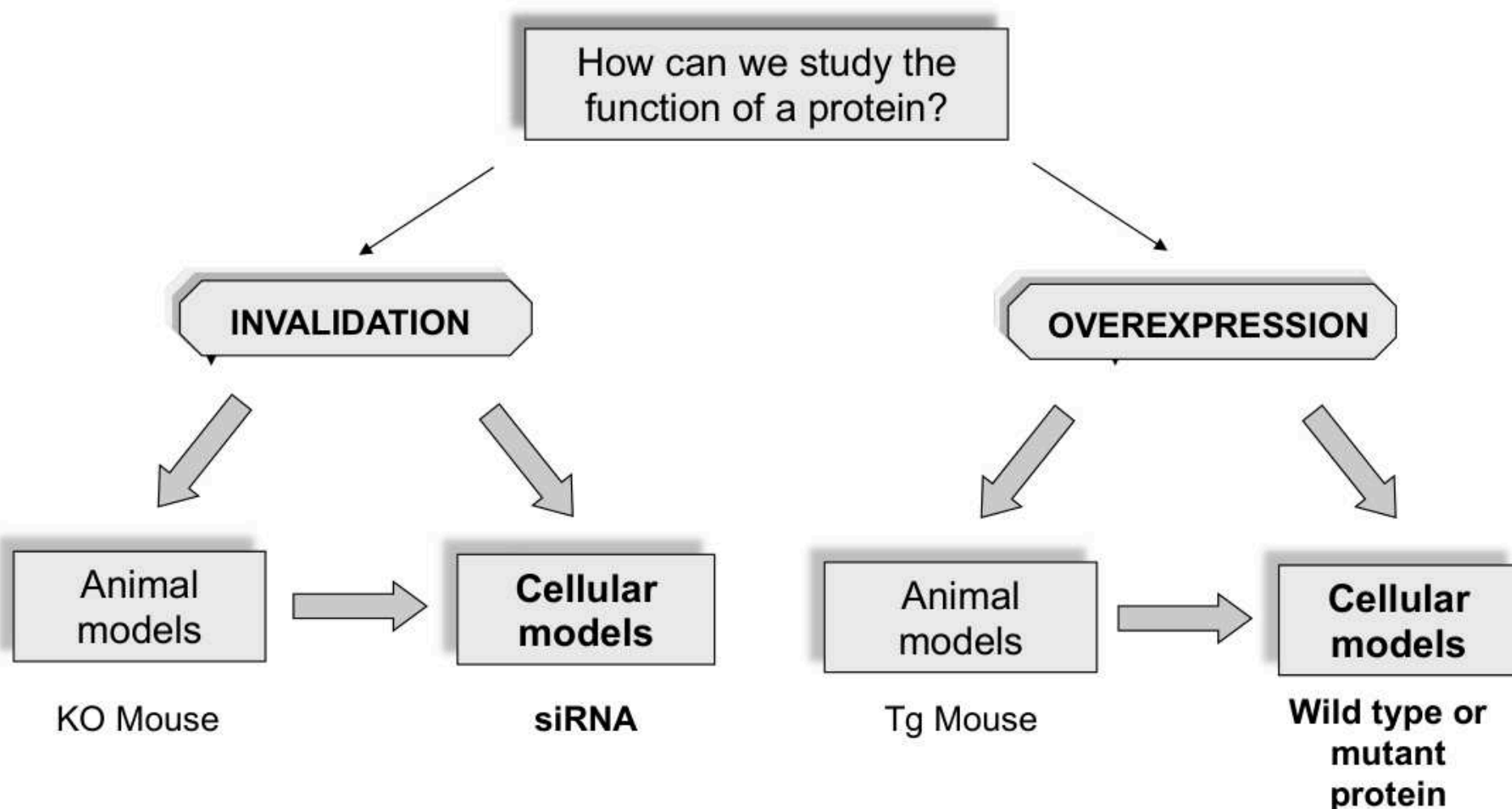
Or cell death



V-2 *In vitro* models- cell lines culture: transfection of cell lines



V-2 *In vitro* models- cell lines culture: transfection of cell lines



V-2 *In vitro* models- cell lines culture: transfection of cell lines

How?

V-2 *In vitro* models- cell lines culture: transfection of cell lines

What construction should be carried out? What vector should be used?

a- Choice of host cell

appropriate cell type

V-2 *In vitro* models- cell lines culture: transfection of cell lines

What construction should be carried out? What vector should be used?

a- Choice of host cell

appropriate cell type

b- Choice of DNA fragment

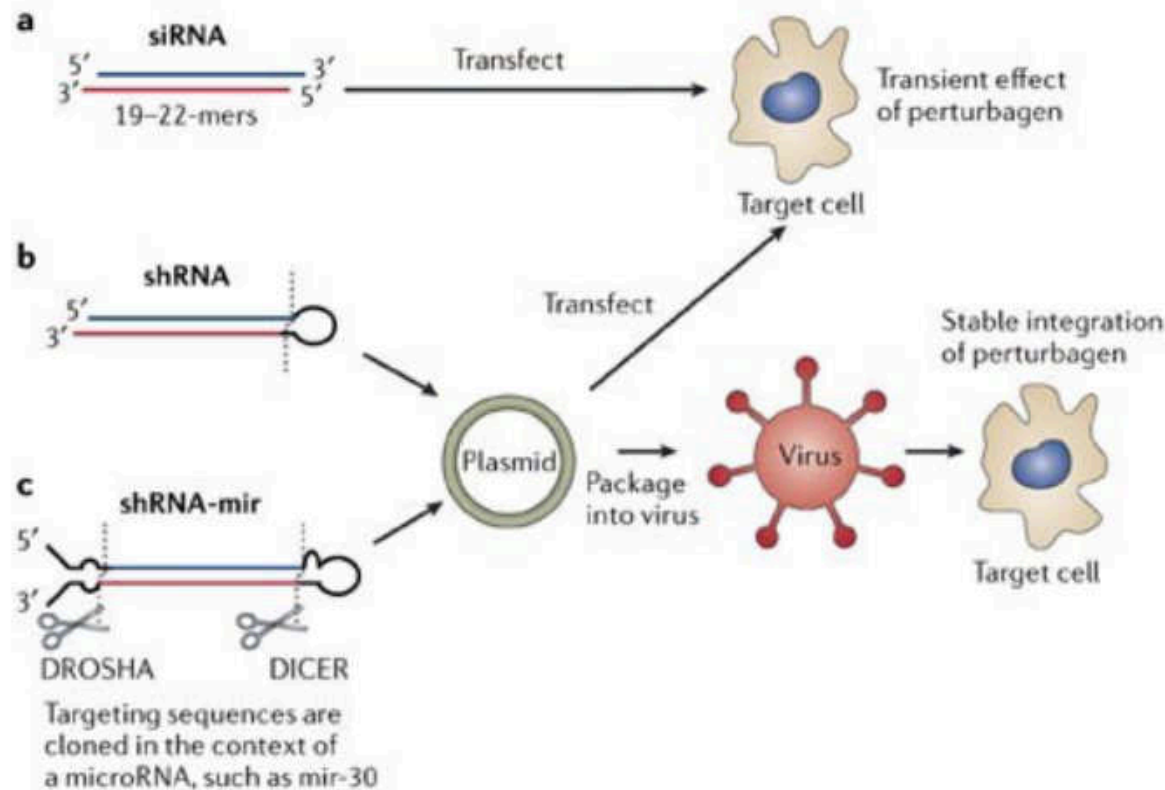
overexpression of wild-type or mutant protein

siRNA-mediated knockdown

V-2 *In vitro* models- cell lines culture: transfection of cell lines

What construction should be carried out? What vector should be used?

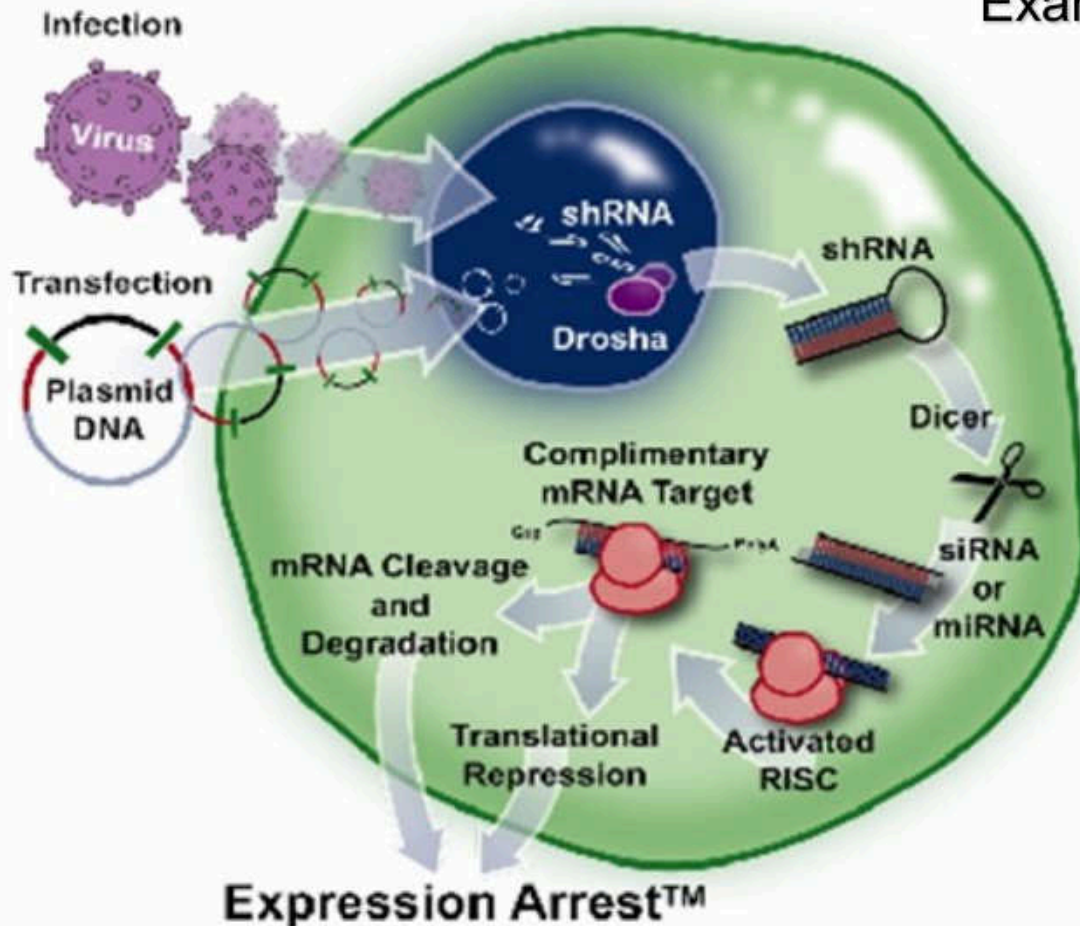
Interfering RNAs



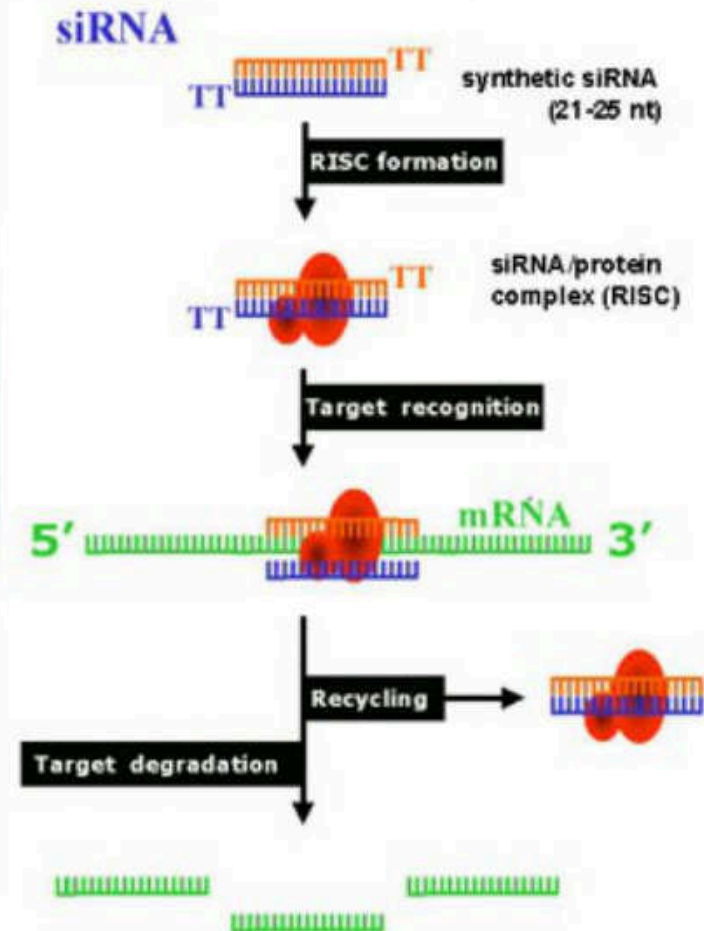
V-2 In vitro models- cell lines culture: transfection of cell lines

What construction should be carried out? What vector should be used?

3. Cell transfection



Example of invalidation by shRNA



RISC: RNA induced silencing complex

V-2 *In vitro* models- cell lines culture: transfection of cell lines

What construction should be carried out? What vector should be used?

a- Choice of host cell

appropriate cell type

b- Choice of DNA fragment

overexpression of wild-type or mutant protein

siRNA-mediated knockdown

c- Choice of vector

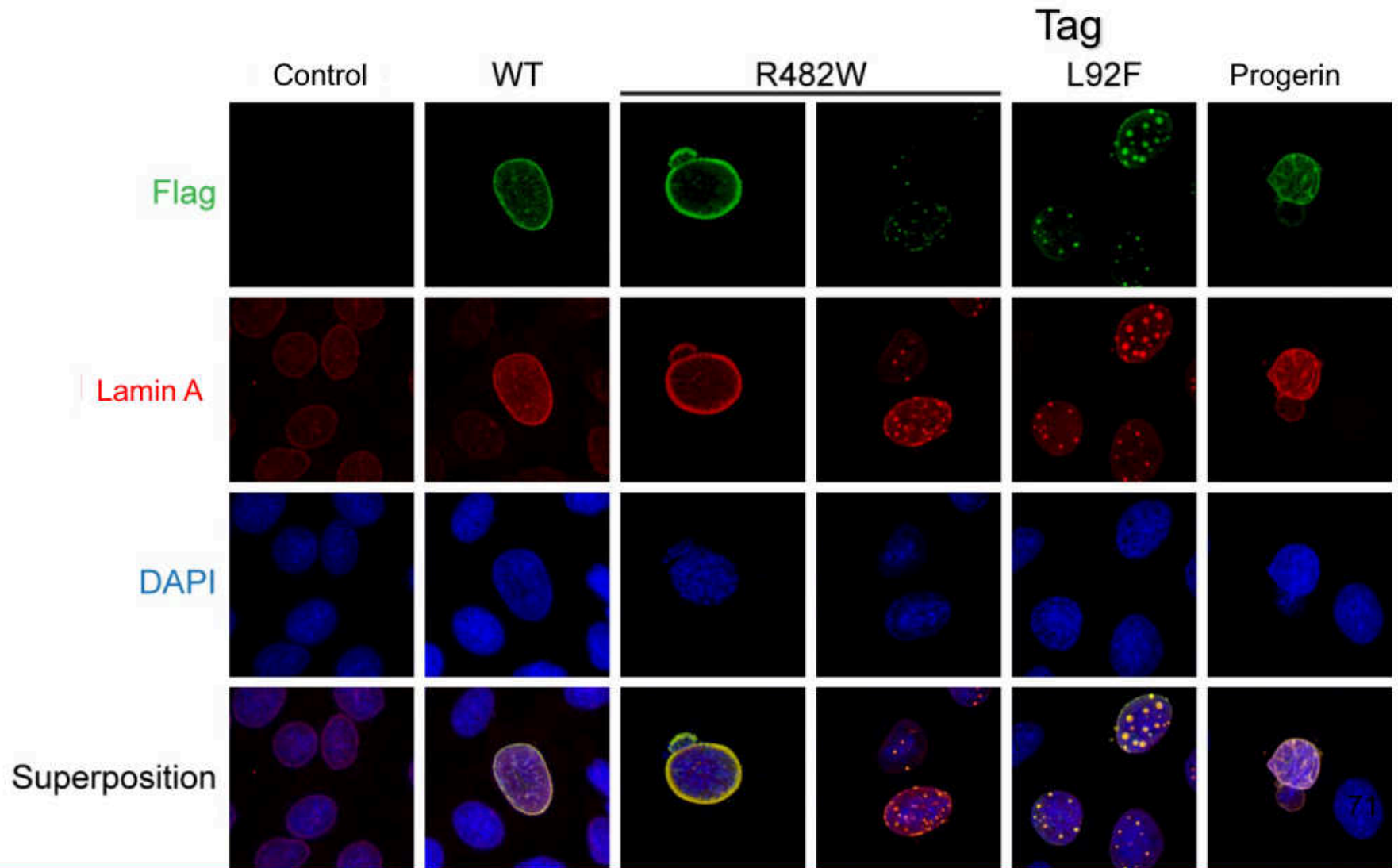
classical plasmids

viruses (adenovirus, lentivirus, retrovirus)

But also choice of promoter, and of a tracking system

V-2 *In vitro* models- cell lines culture: transfection of cell lines

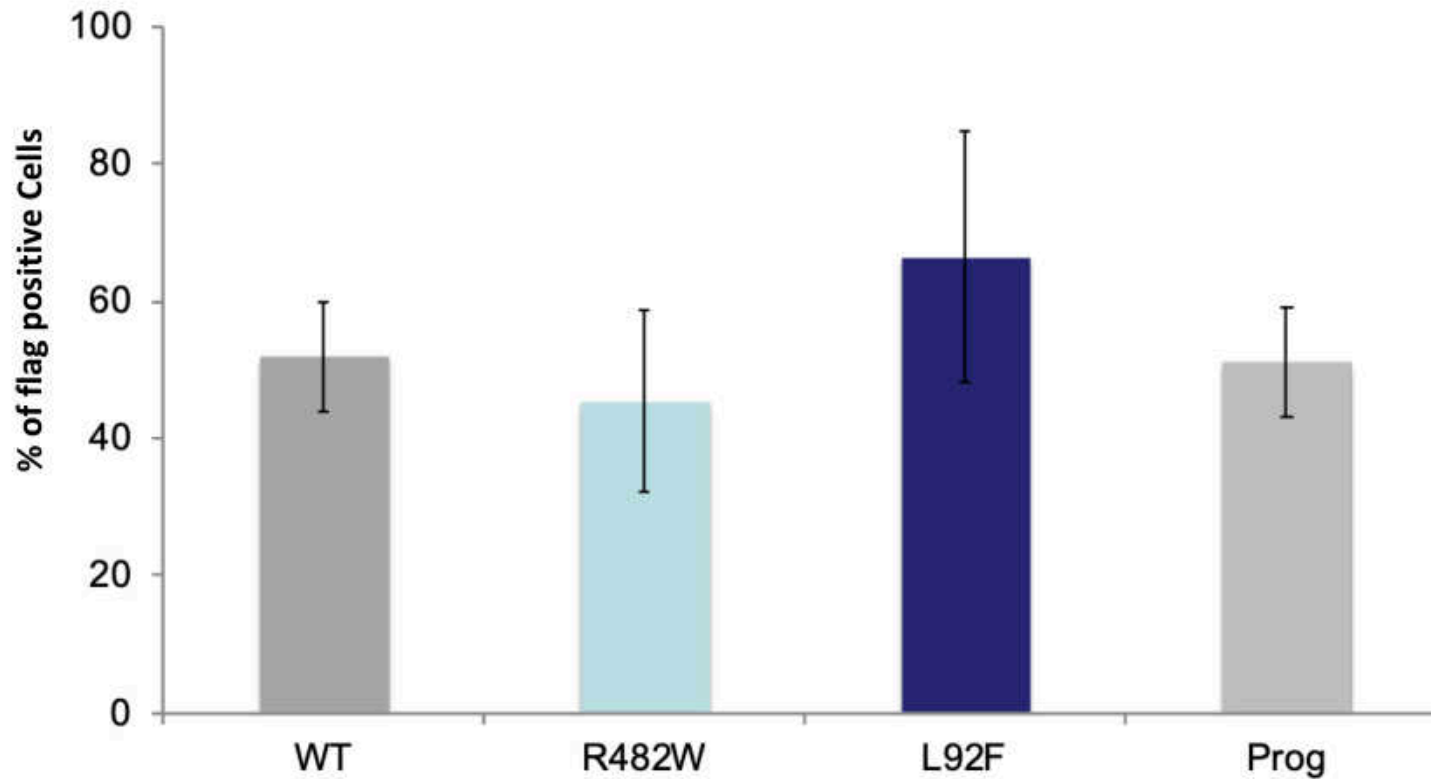
What construction should be carried out? What vector should be used?



V-2 *In vitro* models- cell lines culture: transfection of cell lines

What construction should be carried out? What vector should be used?

Mesure of transfection efficiency



V-2 *In vitro* models- cell lines culture: transfection of cell lines

What construction should be carried out? What vector should be used?

a- Choice of host cell

appropriate cell type

b- Choice of DNA fragment

overexpression of wild-type or mutant protein

siRNA-mediated knockdown

c- Choice of vector

classical plasmids

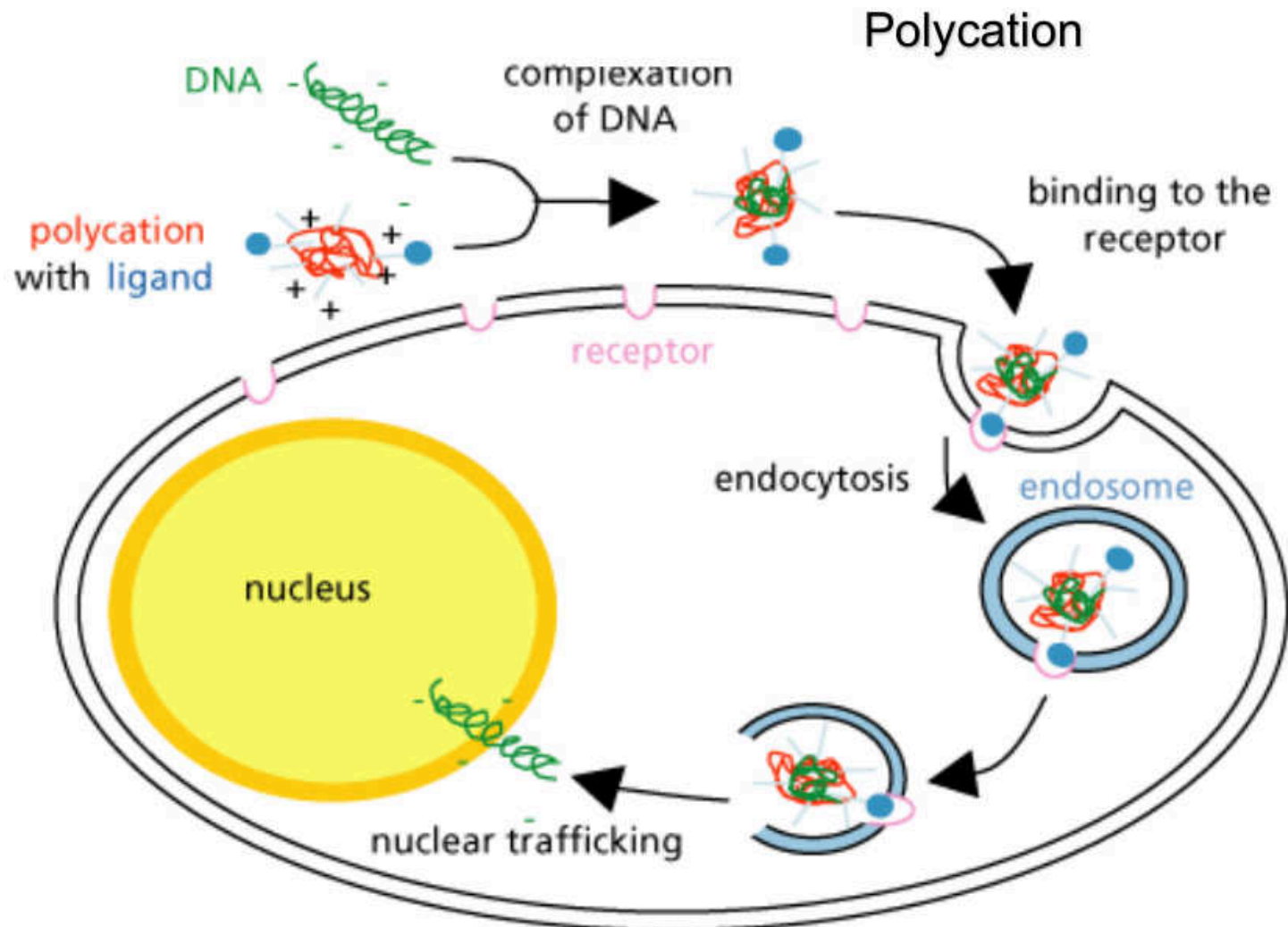
viruses (adenovirus, lentivirus, retrovirus)

But also choice of promoter, and of a tracking system

d- Choice of transfection protocol

V-2 *In vitro* models- cell lines culture: transfection of cell lines

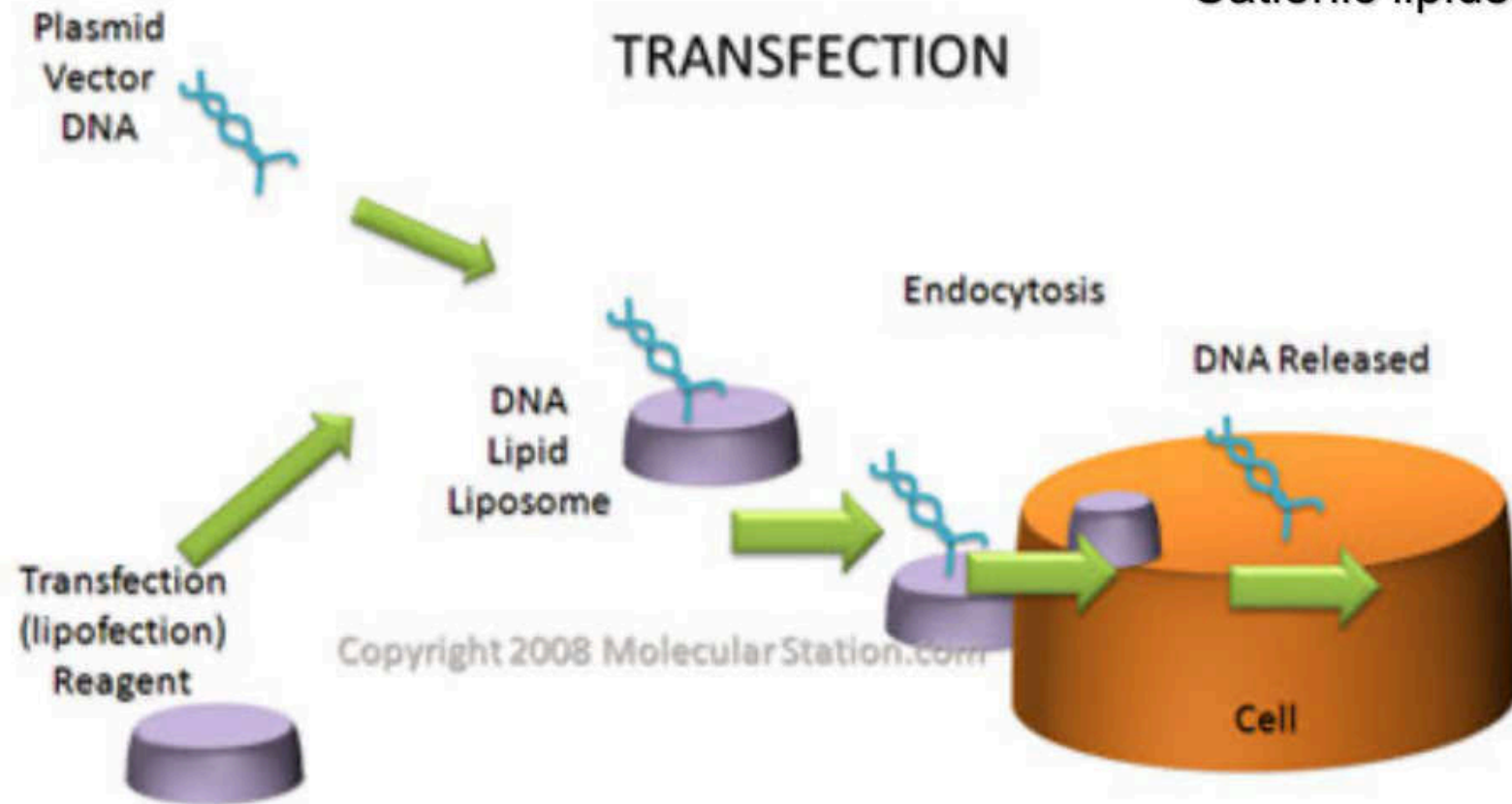
What construction should be carried out? What vector should be used?



V-2 *In vitro* models- cell lines culture: transfection of cell lines

What construction should be carried out? What vector should be used?

Cationic lipids

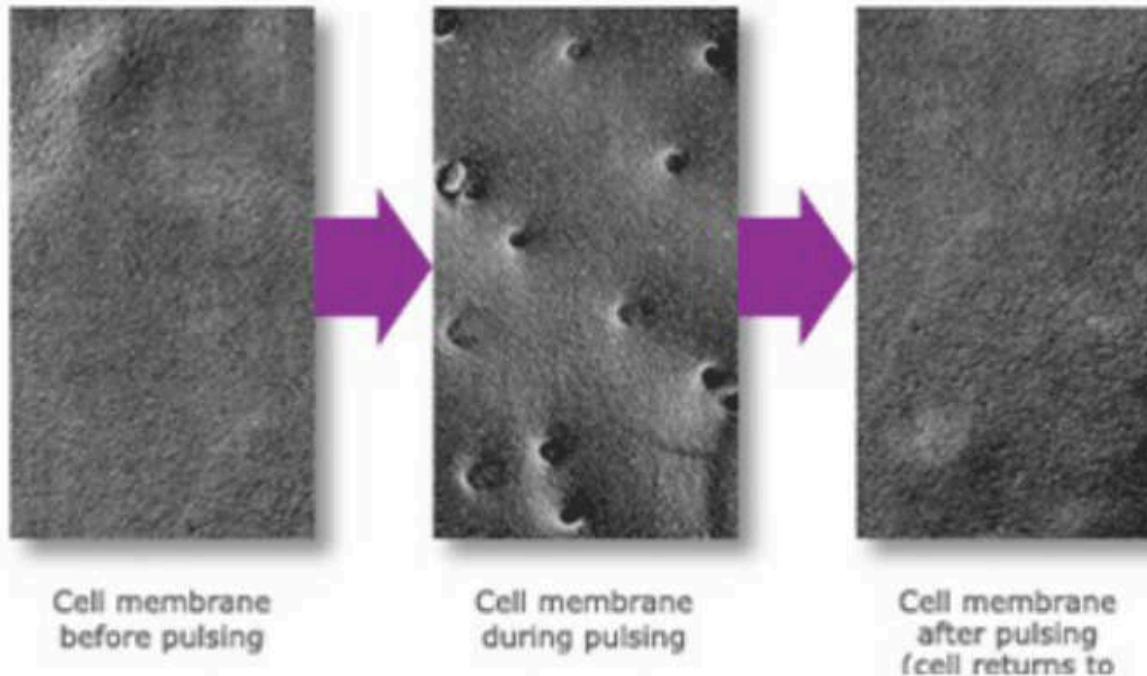


V-2 *In vitro* models- cell lines culture: transfection of cell lines

What construction should be carried out? What vector should be used?

Electroporation

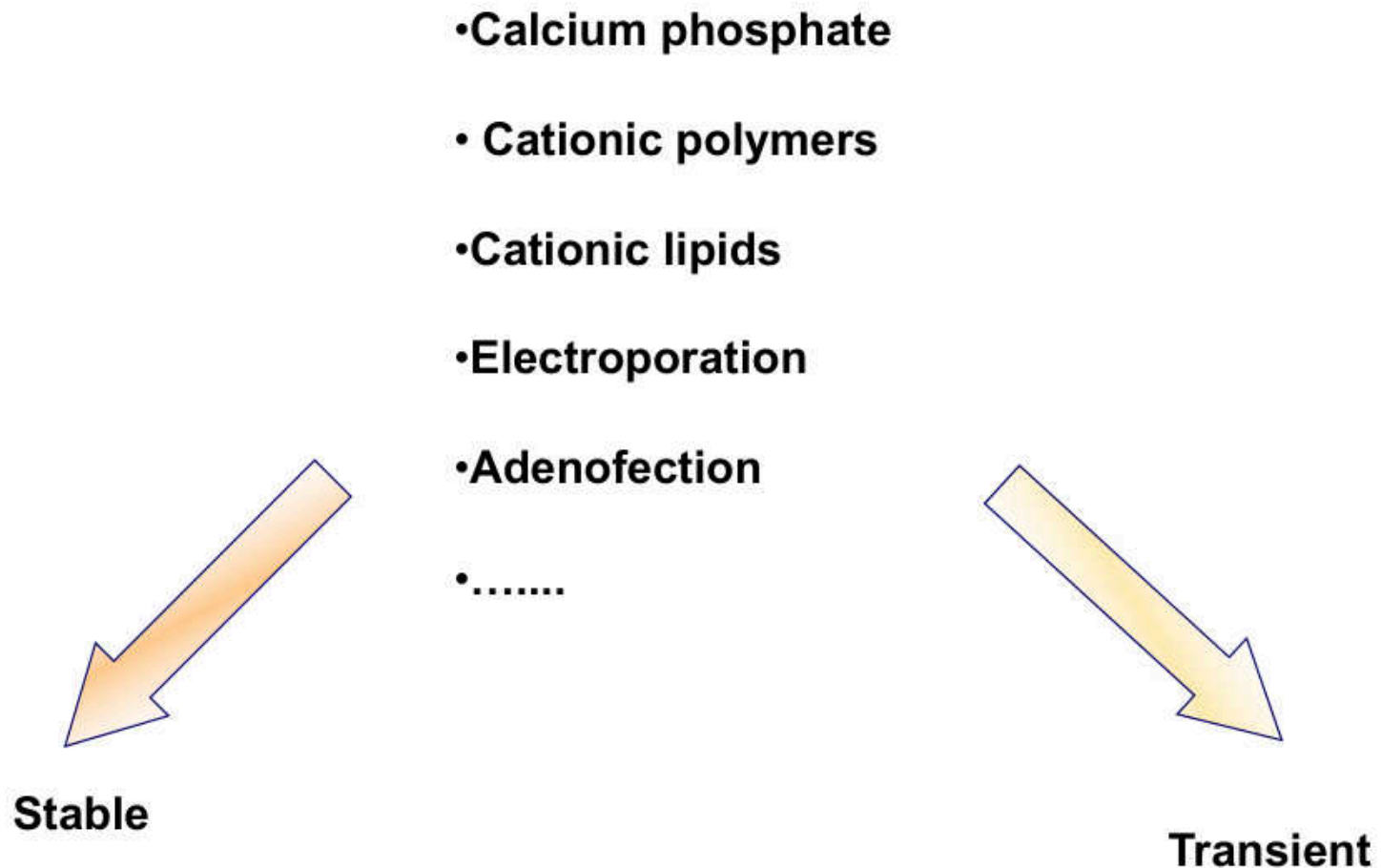
The phenomenon of electroporation



- *Controlled, millisecond electrical pulses induce temporary pores in the cell membrane*
- *Cell membrane reseals and is left unharmed*

V-2 *In vitro* models- cell lines culture: transfection of cell lines

What construction should be carried out? What vector should be used?



A- Mechanisms involved in weight gain and adipose tissue damage mediated by new antiretroviral drugs in people living with HIV

B- Impact of adipose tissue aging on the development of atherosclerosis

C- Impact of progerin in cardio-metabolic complications in patients with progeria?

HUTCHINSON-GILFORD PROGERIA SYNDROME



<http://www.progeriaresearch.org/>

Face :	Facial dysmorphia Prominent eyes pinched nose
Growth :	Growth retardation Weight loss
Skin/bone :	Generalized alopecia Micrognathia

Rare Disease : 1/20 000 000

With the progression of the disease

Accelerated aging syndrome

Skin/bone :	Thinning of the skin Osteolysis of the clavicles and phalanges
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Loss of subcutaneous adipose tissue

Insulin resistance

Cardiovascular complications: Early atherosclerosis

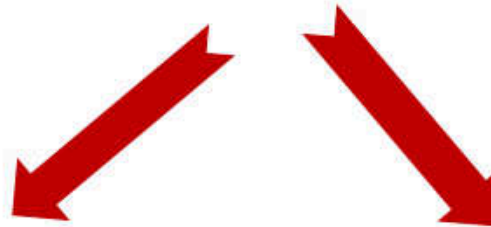
Died of heart failure/stroke at around age 15

Abnormal accumulation of progerin

**Hutchinson-
Gilford
Progeria
Syndrome
(HGPS)**

- Mutation p.G608G
- Progerin accumulation

**Accelerated cellular/tissue aging
Linked to progerin**



**(1) Study of AT
damage**

**(2) Study of cardio-metabolic
complications**

= Exacerbated age-related conditions



V Béréziate PU

HGPS

Accumulation of Progerin

?

Atherosclerosis



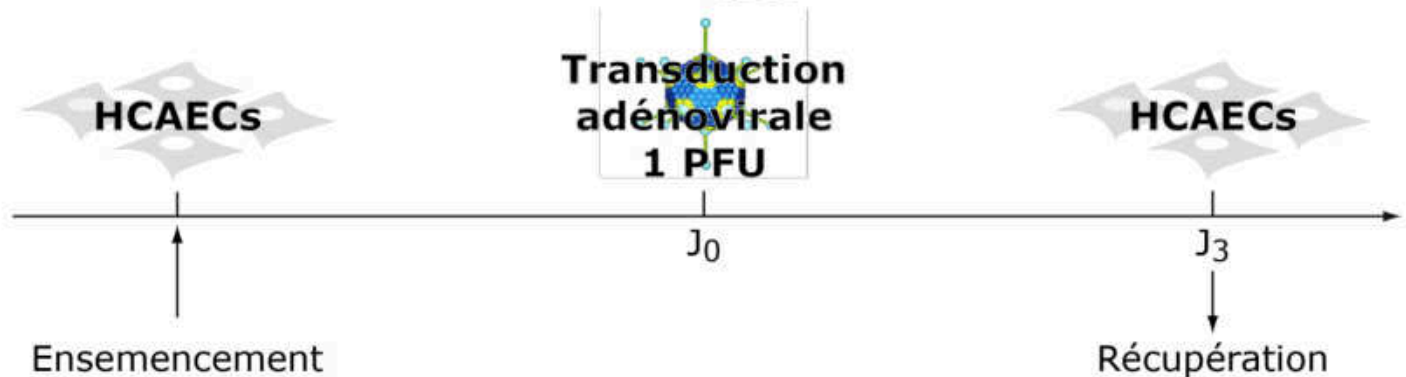
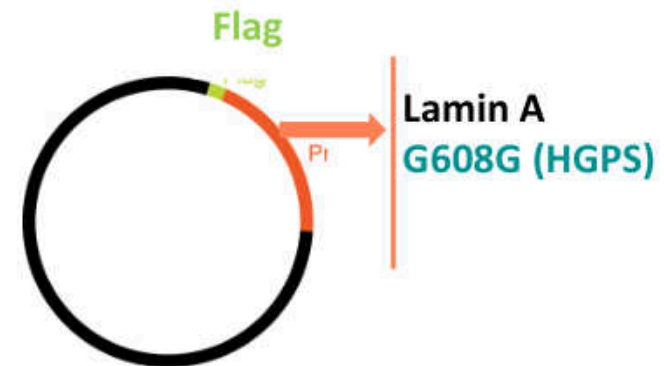
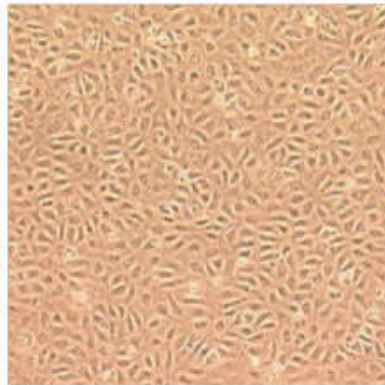
Guillaume Bidault
Master BIP, Ph.D

Postdoctoral fellowship in Cambridge
Research associate in Institute of Metabolic Science

HGPS: Hutchinson-Gilford Progeria Syndrome

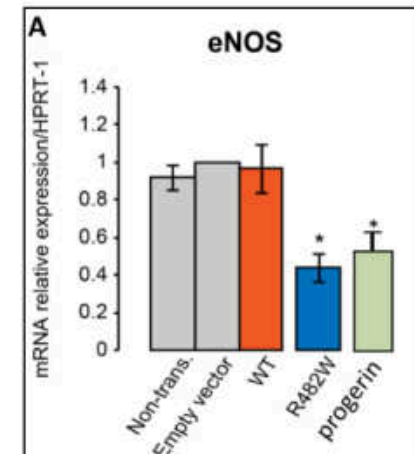
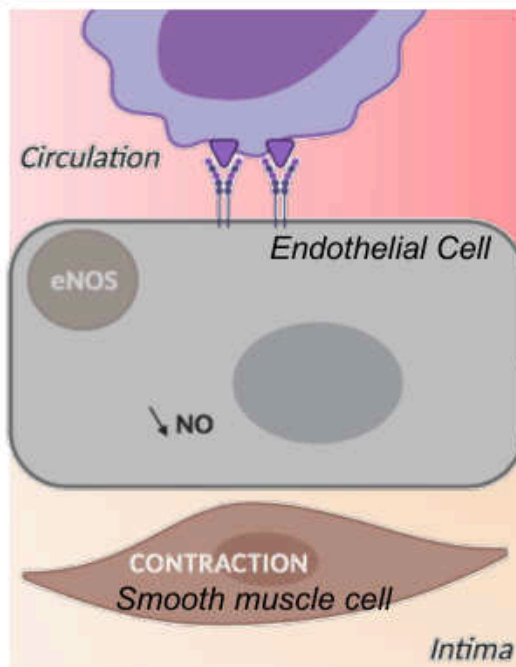
Primary culture of endothelial cells derived from transfected human coronary arteries (HCAECs)

IN VITRO



p.G608G (HGPS)- p.R482W (FPLD2)

Endothelial dysfunction

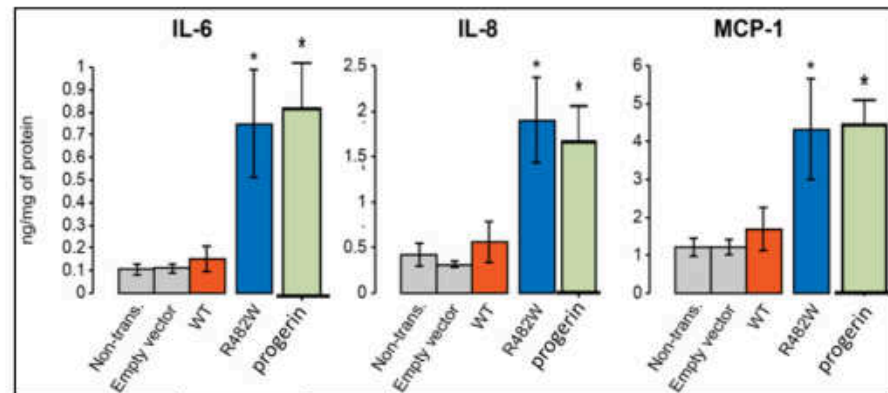
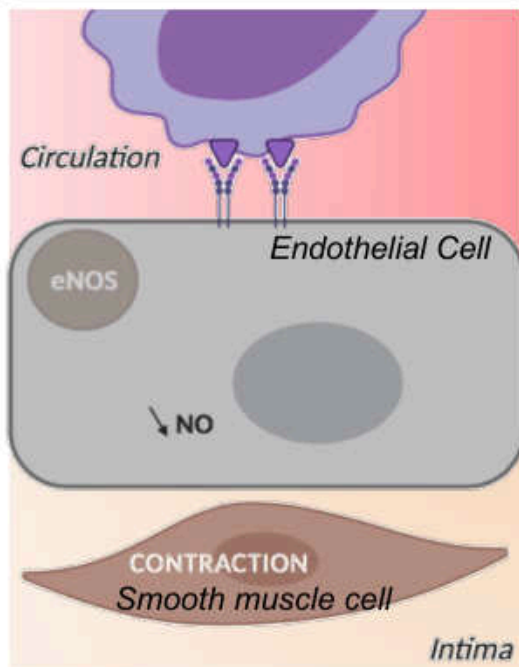


1. ➔ Endothelial NO synthase Expression

HGPS: Hutchinson-Gilford Progeria Syndrome
FPLD2: Familial Partial Lipodystrophy of Dunnigan

NO: Nitric Oxide
eNOS: endothelial NO synthase
CAM: Cellular Adhesion Molécules

p.G608G (HGPS)- p.R482W (FPLD2)
Endothelial dysfunction



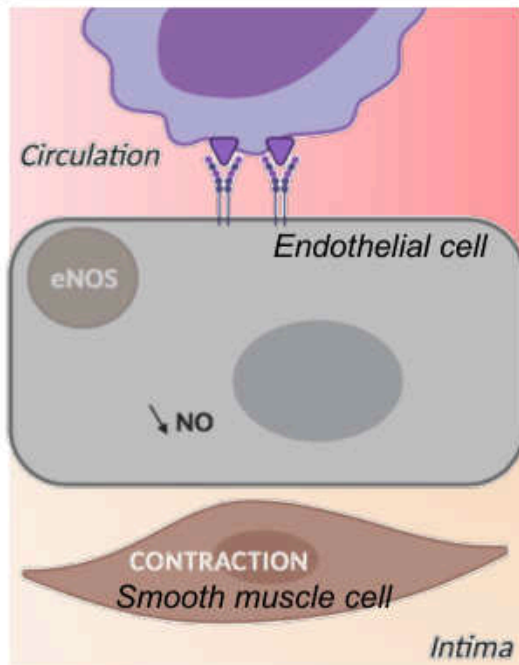
2. ↗ Inflammation

HGPS: Hutchinson-Gilford Progeria Syndrome
 FPLD2: Familial Partial Lipodystrophy of Dunnigan

NO: Nitric Oxide
 eNOS: endothelial NO synthase
 CAM: Cellular Adhesion Molécules

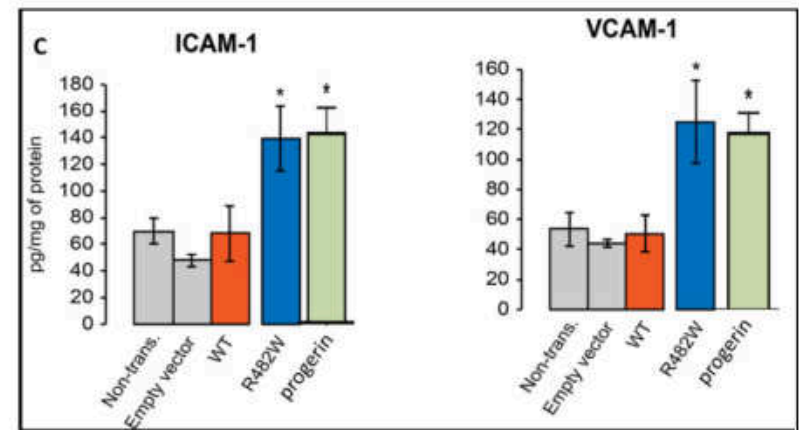
p.G608G (HGPS)- p.R482W (FPLD2)

Endothelial dysfunction



HGPS: Hutchinson-Gilford Progeria Syndrome
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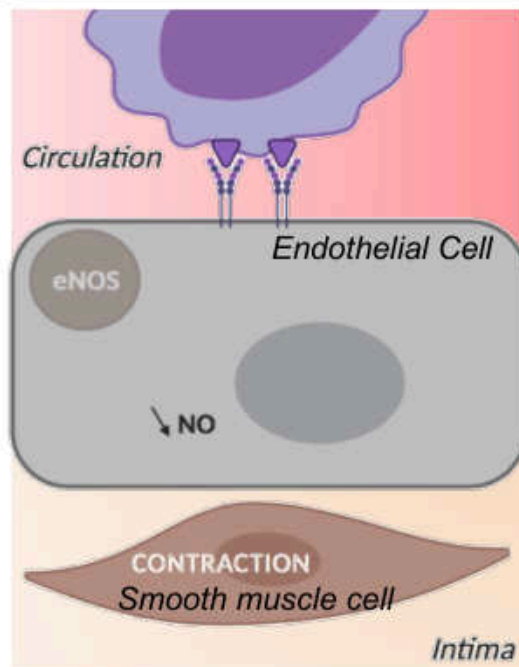
3. Adhesion Molecules ICAM, VCAM expression



NO: Nitric Oxide
eNOS: endothelial NO synthase
CAM: Cellular Adhesion Molecules

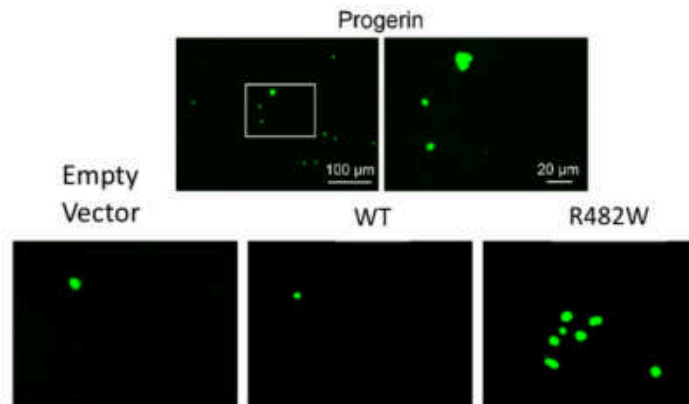
p.G608G (HGPS)- p.R482W (FPLD2)

Endothelial dysfunction



4. Monocytes adhesion

D

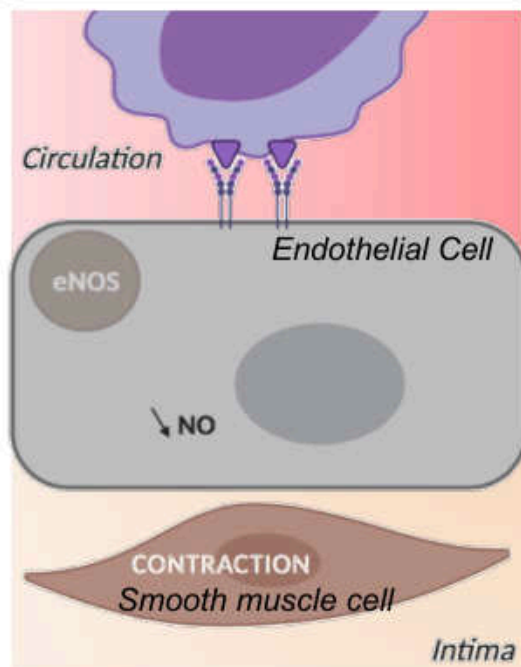


Adherent PBMCs to endothelial cells:
green staining

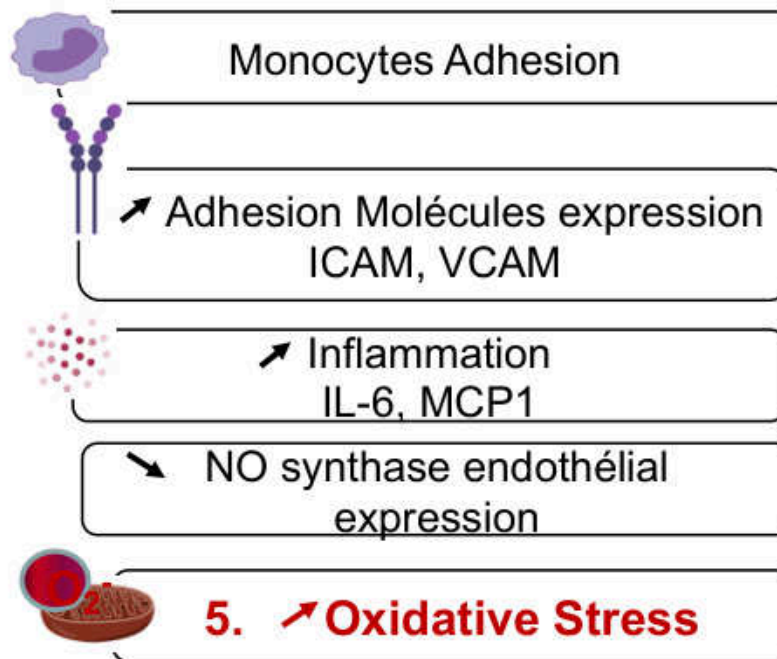
HGPS: Hutchinson-Gilford
Progeria Syndrome
FPLD2: Familial Partial
Lipodystrophy of Dunnigan
PBMC: Peripheral Blood
Mononuclear Cells

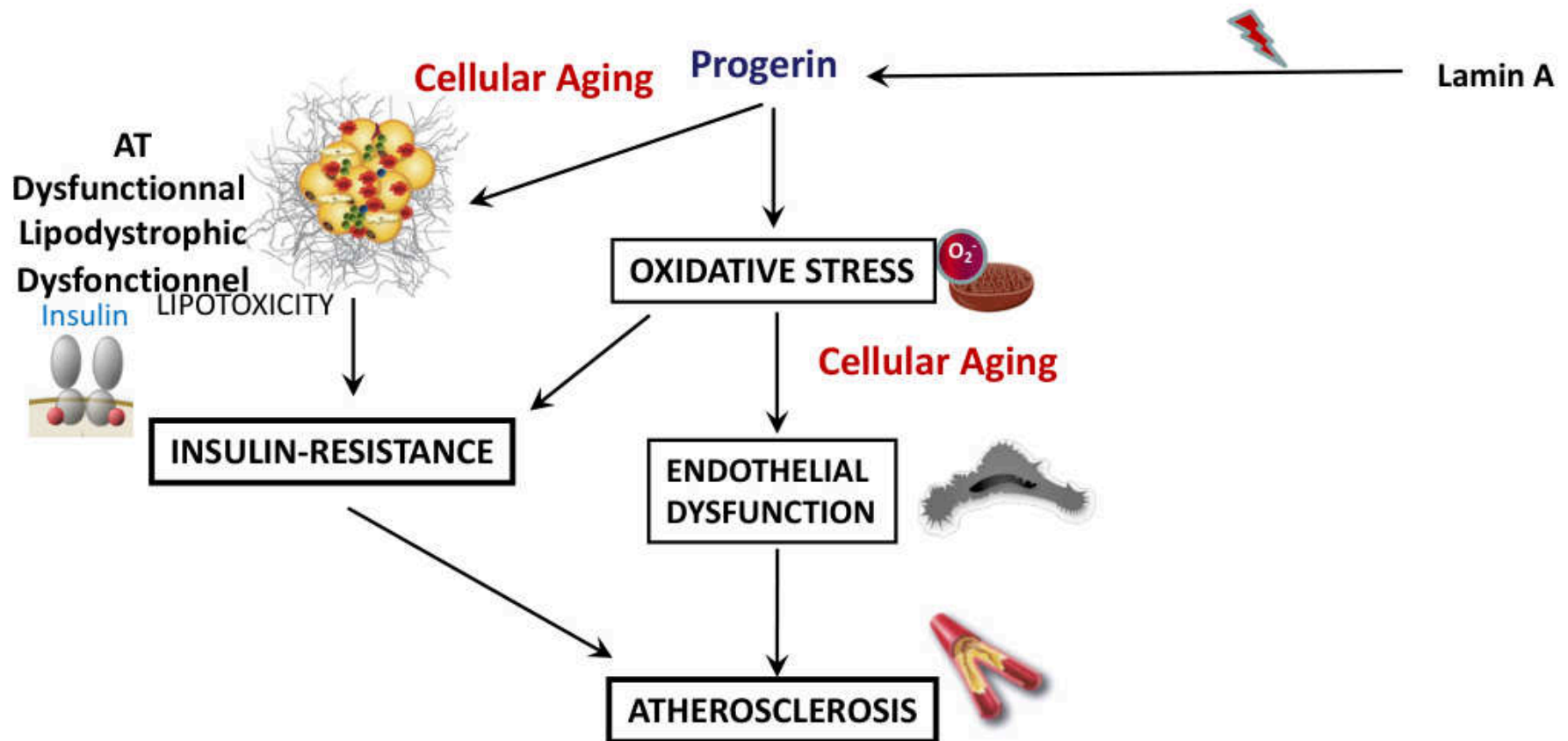
NO: Nitric Oxide
eNOS: endothelial NO synthase
CAM: Cellular Adhesion Molecules

p.G608G (HGPS)- p.R482W (FPLD2)
Endothelial dysfunction



HGPS: Hutchinson-Gilford Progeria Syndrome
 FPLD2: Familial Partial Lipodystrophy of Dunnigan





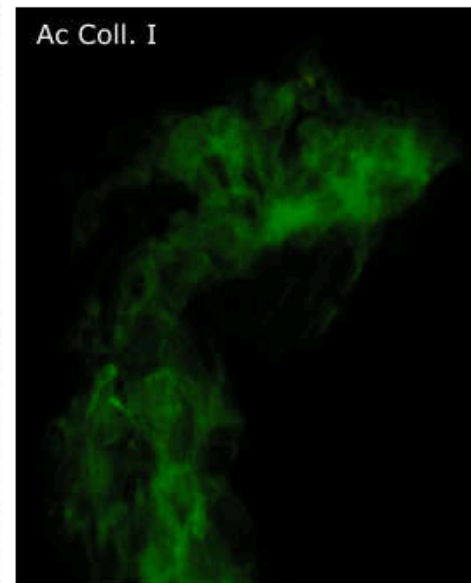
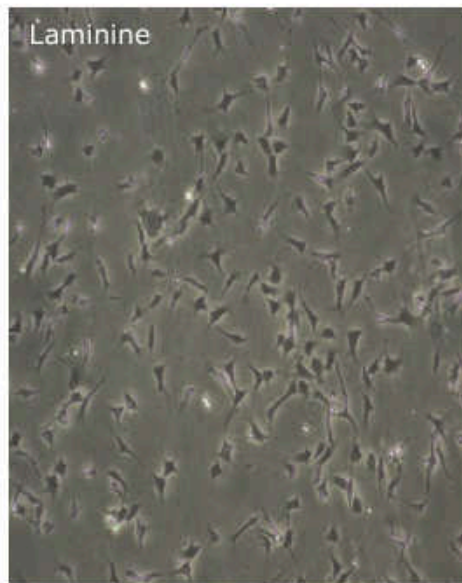
II-2 Type of support

Single-layer culture on plastic substrate



VI-1 Material: Type of support

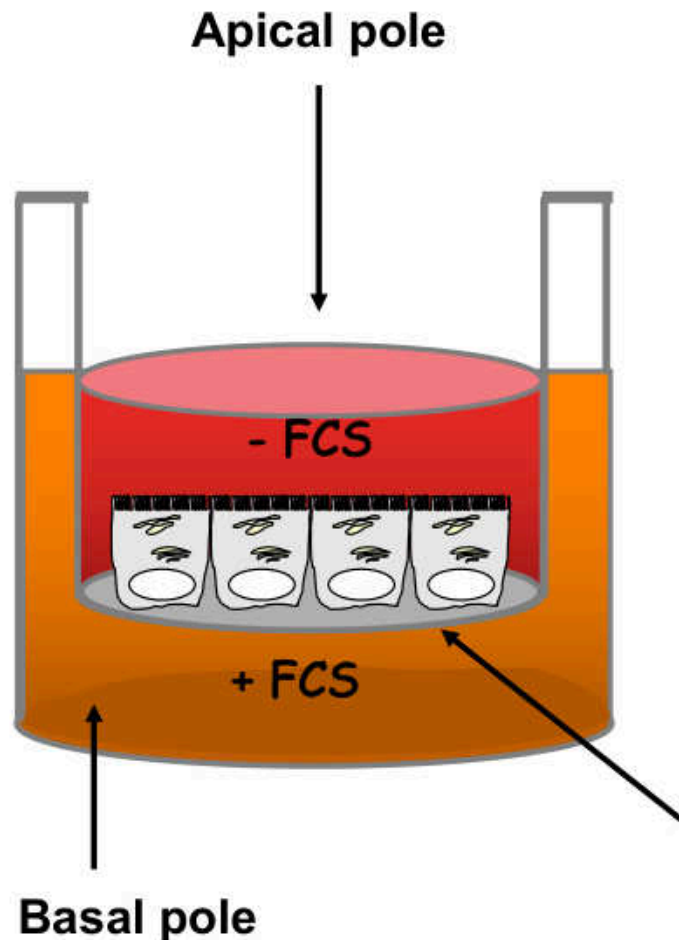
Monolayer culture on matrix protein-coated substrates



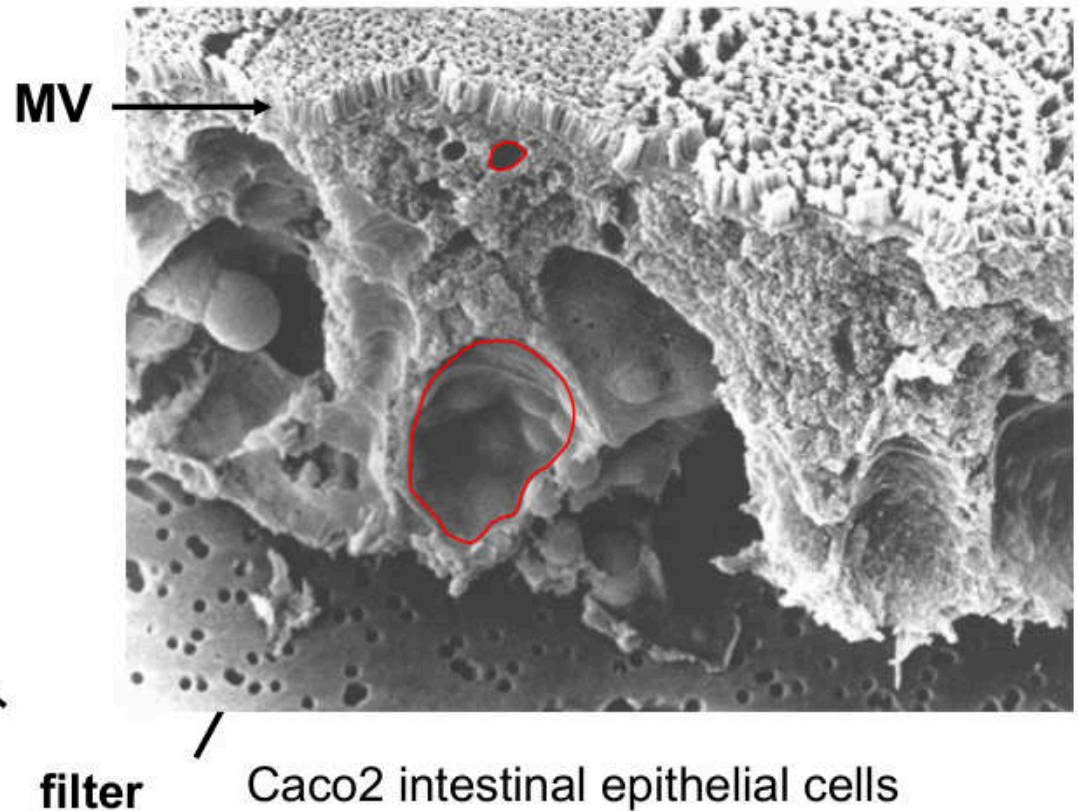
human chondrocytes in primary culture

VI-1 Material: Type of support

Single-layer culture on filter



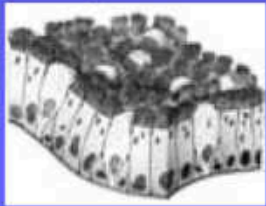
Scanning electron microscopy



Three-dimensional culture

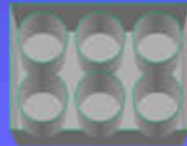
Three-dimensional *in vitro* model of human tracheobronchial epithelium regeneration: spheroids

polypes nasaux ou
trachée foetale
mature



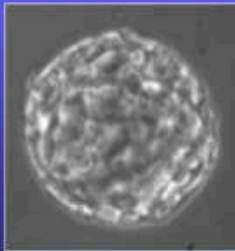
Cellules épithéliales
respiratoires isolées

Milieu de Culture :
Ham's F12 + 2% Ultrosér G.



Rotation permanente
100 rpm

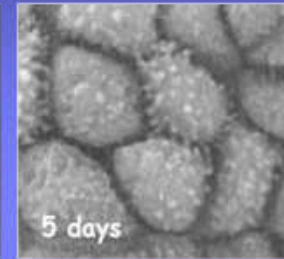
$50 \leq \varnothing \leq 100$
 μm



48h



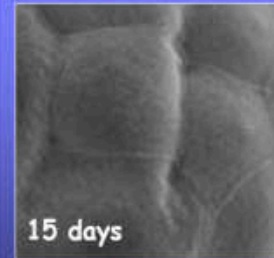
5 days



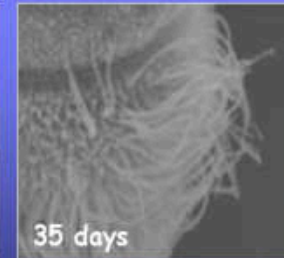
5 days



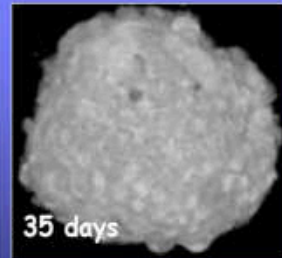
15 days



15 days



35 days



35 days

VI-2 Material: Type of culture room

- **Their pathogenicity** to humans
- **The risk of spread** in the community
- The existence of **prophylactic or curative treatment**

L1-type culture room

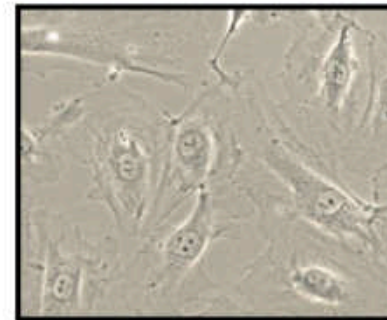
- primary mouse culture
- immortalized murine cell lines
- class 1 OGM



3T3F442A

L2-type culture room

- primary human culture
- class 1 OGM
- some immortalized lineages
- some primary animal culture



HCAECs

• *Biological agents that may pose a risk of disease to humans but are unlikely to spread. Treatments exist to combat this type of disease.*

VI-2 Material: Type of culture room



L2 culture room

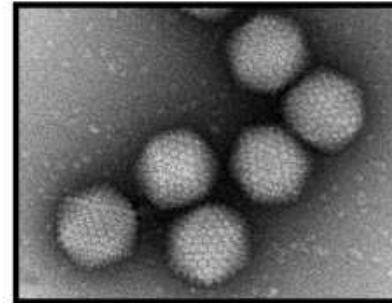


VI-2 Material: Type of culture room

L3-type culture room

- Class 2 or 3 OGM according to HCB
- échantillons biologiques infectés

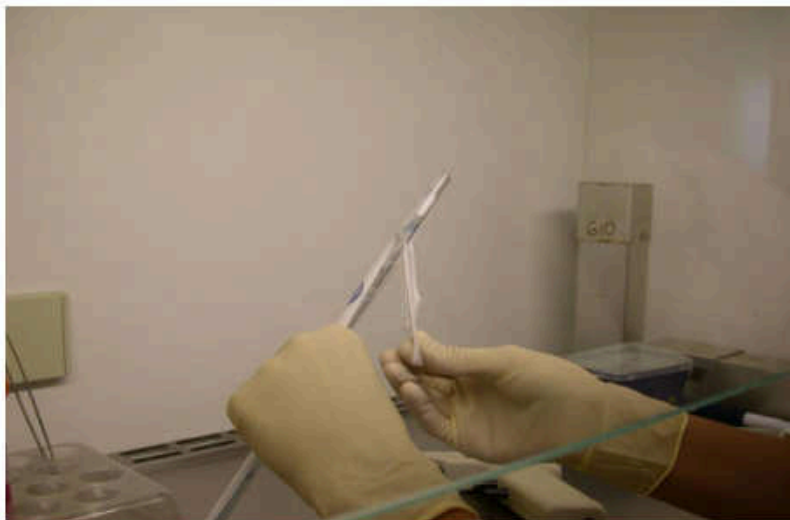
Biological agents are pathogenic to humans and can spread. Treatments generally exist to combat this type of disease.



Production
adénovirale



VI-2 Material: Type of culture room



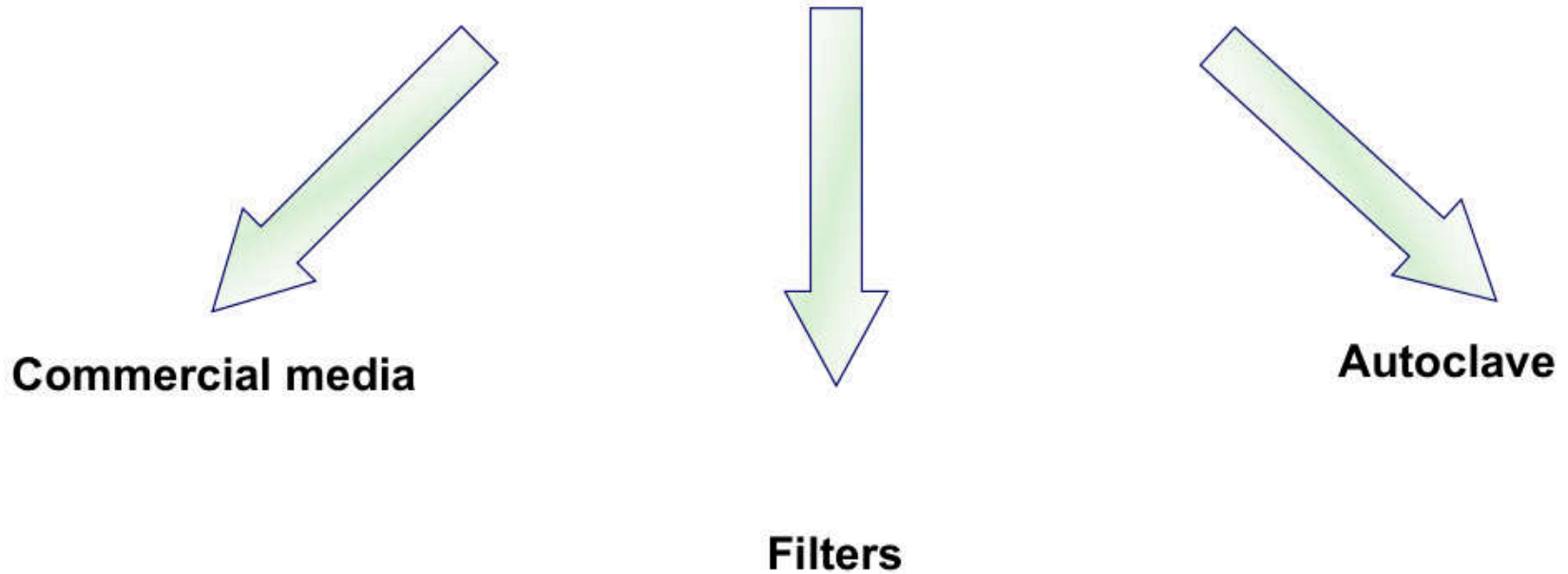
VI-3 Material: Cell media

Culture media must meets the cultural needs of animal cells.

- H₂O
 - minerals
 - source of carbon and energy
 - nitrogen source
 - essential amino acids
 - vitamine
 - Fatty acids
 - Constant pH/buffer system
-
- Antibiotics
 - L-glutamine,
 - Sodium pyruvate
 - Growth factors
 - A reducing agent (such as β -mercaptoethanol))
 - a differentiation-inducing agent



VI-3 Material: Cell media

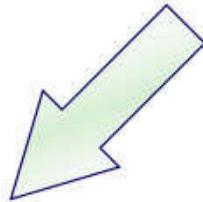


VI-3 Material: Cell media

Source of contamination

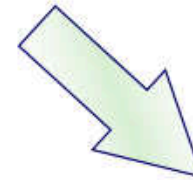
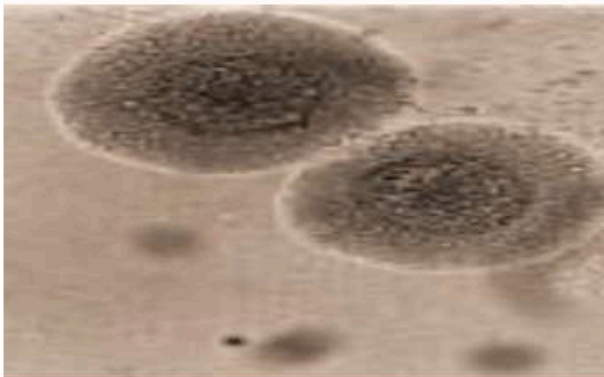
- the origin of cells
- Reagents/materials used in cell culture
- The experimenter

Type of Contamination



Bacteria:

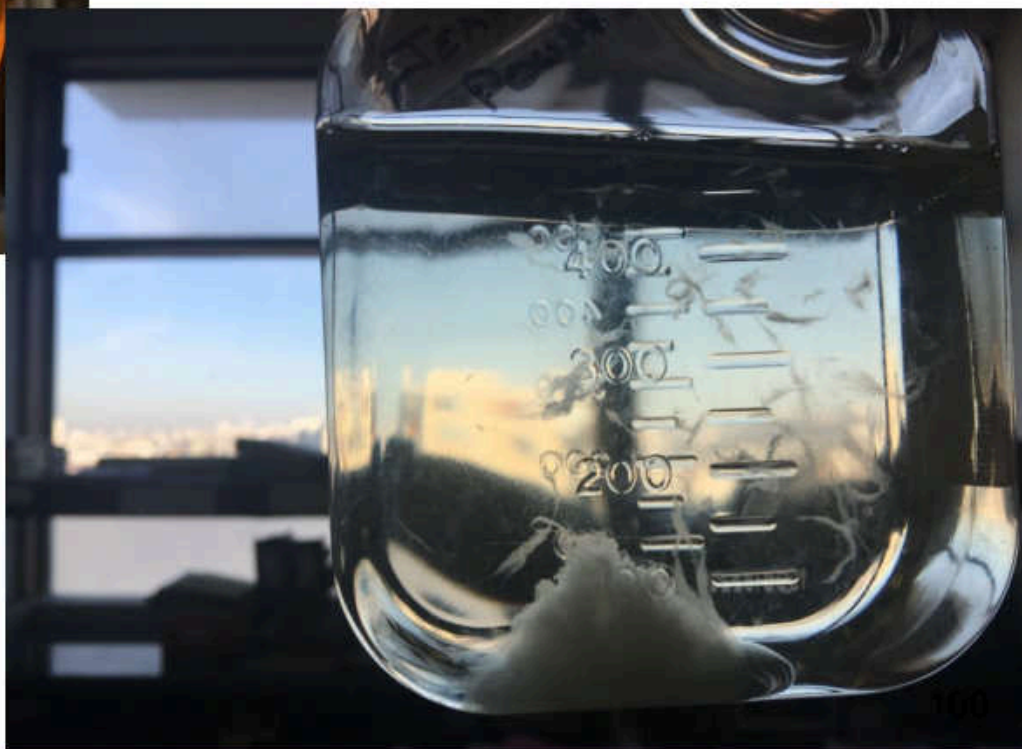
Special case of mycoplasmas



Mushrooms



VI-3 Material: Cell media



advantages and limitations?

